

INVENTION DISCLOSURE

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RRM-Driven, Radio-Aware Automated Frequency Coordination (AFC) Request Management and Enforcement for Multi-Radio Access Points

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ABSTRACT

The present invention discloses a comprehensive, Radio Resource Management (RRM)-driven framework for the complete lifecycle management of Automated Frequency Coordination (AFC) requests and responses in multi-radio, heterogeneous wireless access points (APs) operating in the 6 GHz band (5.925–7.125 GHz). Existing AFC implementations operate as isolated subsystems, lacking coordination with RRM, radio topology awareness, dynamic provider selection, or intelligent response caching. This invention introduces a novel distributed Storm-bolt topology—comprising RrmAFCBolt, RrmHandlerBolt, and RrmACSBolt—that governs AFC from initial GPS-based geolocation validation through request payload construction, multi-provider dynamic routing, GET-before-POST query optimisation, response parsing, EIRP/PSD enforcement, 3-tier expiry-aware cache management, and atomic AP lifecycle (claim/unclaim) workflows. Key innovations include: (1) ECEF-based AP location-change detection with configurable displacement thresholds; (2) PSD-to-EIRP dual-source computation with AFC_MIN_PSD bandwidth floors; (3) 320 MHz Wi-Fi 7 Mode-1/Mode-2 channel disambiguation; (4) LPI fallback for GPS-unavailable standard-power APs; (5) per-site, policy-driven, dynamic AFC provider selection; (6) radio-aware AFC origination from 2.4 GHz radio interfaces when dedicated 6 GHz radios are absent; and (7) AP outage recovery with AFC re-authorisation. The framework ensures regulatory compliance under FCC 47 CFR Part 15 Subpart E (US) and ISED RSS-248 (Canada) while maximising spectrum utilisation and network resilience.

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1. TECHNICAL FIELD

The present invention relates to wireless local area networks (WLANs) and, more particularly, to Radio Resource Management (RRM)-driven lifecycle control of Automated Frequency Coordination (AFC) in heterogeneous multi-radio Access Points (APs) operating across the 2.4 GHz, 5 GHz, and 6 GHz frequency bands under the IEEE 802.11ax (Wi-Fi 6E) and IEEE 802.11be (Wi-Fi 7) specifications.

More specifically, the invention relates to the complete AFC workflow encompassing: geolocation acquisition and validation; AFC request payload construction conformant to WINNF-TS-1014 (AFC System to AFC Device Interface Specification v1.4); dynamic AFC provider selection and redirection; GET-before-POST query optimisation; AFC response parsing and per-channel Equivalent Isotropic Radiated Power (EIRP) enforcement; Power Spectral Density (PSD)-to-EIRP conversion; bandwidth-specific AFC Minimum PSD floor enforcement; 320 MHz Wi-Fi 7 channel mode disambiguation; response expiry and cache lifecycle management; AP claim and unclaim workflows; Low Power Indoor (LPI) fallback for GPS-unavailable Standard Power (SP) APs; and outage-recovery re-authorisation—all governed by RRM control logic within a distributed Apache Storm stream processing topology deployed on the Mist Cloud Platform (Juniper Networks / Hewlett Packard Enterprise).

The invention is specifically directed to enabling regulatory compliance with: FCC 47 CFR Part 15, Subpart E (United States); ISSED RSS-248, Issue 1 (Canada); and ITU-R M.2116 technical parameters for AFC systems in the 5.925–7.125 GHz frequency range.

2. BACKGROUND AND PROBLEM STATEMENT

2.1 Regulatory Context and Prior AFC Implementations

The 6 GHz frequency band (5.925–7.125 GHz) has been opened for unlicensed Wi-Fi use in the United States under FCC Report and Order FCC-20-51, and subsequently in Canada, the United Kingdom, Brazil, Chile, and numerous other jurisdictions. Unlike prior unlicensed bands, the 6 GHz band contains incumbent users (fixed microwave links, broadband service providers, satellite earth stations, and cable TV relay systems) that must be protected from interference. The FCC has mandated that Standard Power (SP) devices—capable of operating at up to 36 dBm EIRP—must not transmit without first obtaining authorisation from an Automated Frequency Coordination (AFC) system.

The AFC system, as defined by the Wi-Fi Alliance (WFA) and WINNF-TS-1014 specification, operates as a cloud-based spectrum database that accepts device registration requests containing the device's geolocation, certification credentials, and inquired frequency ranges. The AFC system returns available channels with per-channel maximum EIRP and Power Spectral Density (PSD) limits, along with an authorisation expiry timestamp. Devices must re-register upon expiry, upon physical relocation, or upon specific regulatory events.

Prior AFC implementations have focused exclusively on the registration-response cycle at the device or access-point level, treating AFC as a simple pre-transmission authorisation check. These implementations suffer from significant deficiencies when deployed at scale in enterprise and carrier-grade environments with heterogeneous AP hardware, dynamic AP provisioning, and complex radio topologies.

2.2 Deficiency P1: Monolithic, Static AFC Request Handling

Existing AFC implementations construct request payloads using static, pre-configured device parameters—specifically the FCC ID and ruleset identifier—that are embedded in firmware or operating system configuration files. This approach is inadequate for enterprise deployments where AP hardware models change frequently, firmware updates may alter certification identifiers, and individual APs within a site may have different certification credentials based on their manufacturing batch, country of deployment, or hardware revision.

Furthermore, prior implementations treat the AFC request as a one-time, atomic operation without provisions for: (a) dynamic retry upon receiving a RETRY state from the AFC provider; (b) re-submission upon GPS location change detection; (c) response validation using a GET-before-POST protocol to leverage cached provider entries; or (d) provider failover when the primary AFC system is unavailable or returns invalid responses. The absence of these mechanisms results in APs either operating without valid AFC authorisation or remaining unnecessarily offline awaiting manual intervention.

2.3 Deficiency P2: Absence of RRM-AFC Co-ordination

Radio Resource Management (RRM) is responsible for dynamic channel selection, transmit power optimisation, automatic channel selection (ACS), radar avoidance (DFS), and co-channel interference mitigation across all APs within a managed network. In prior art, AFC and RRM operate as entirely independent subsystems: AFC authorises a set of channels and power limits, and RRM separately selects channels from its own interference model without awareness of AFC constraints. This architectural separation causes three critical problems:

- Channel Misalignment: RRM may select a channel that is not in the AFC-authorized set, requiring the AP to operate at reduced power or remain offline until the conflict is resolved.
- EIRP Overestimation: RRM applies generic per-model power tables that may exceed the AFC-determined per-channel EIRP limit, causing illegal transmissions or triggering receiver-protection enforcement.
- Bandwidth Mismatch: AFC responses provide per-channel authorisation for specific bandwidths (20/40/80/160/320 MHz), but RRM's bandwidth selection algorithm is unaware of AFC bandwidth constraints, potentially selecting an unauthorised channel width.

The present invention eliminates these deficiencies by fully integrating AFC lifecycle management within the RRM control plane, ensuring that channel selection, EIRP assignment, and bandwidth decisions are all constrained by real-time AFC authorisation.

2.4 Deficiency P3: Multi-Radio Heterogeneity and Band Targeting

Modern enterprise APs incorporate two, three, or four independent radio interfaces operating across 2.4 GHz, 5 GHz, and 6 GHz bands. Tri-band APs (e.g., AP43, AP45, AP64 in the HPE Aruba / Mist product lines) contain: a dedicated 2.4 GHz radio (r24), a dedicated 5 GHz radio (r5), and a dedicated 6 GHz standard-power radio (r6). Dual-band APs with 6 GHz capability (e.g., AP32E, AP34) contain a shared radio capable of operating in either the 2.4 GHz or 6 GHz band, determined by auto-band assignment.

Prior AFC implementations assume a fixed, dedicated 6 GHz radio and always originate AFC requests from that interface. This assumption fails in the following scenarios:

- **Auto-Band Conversion:** When a 2.4 GHz radio (r24) is configured for auto-band operation (auto_band=True), it may be re-assigned to serve the 6 GHz band. In this case, the AFC request must be originated from the 2.4 GHz radio interface, not a non-existent dedicated 6 GHz interface.
- **Tri-Band AFC Origination:** In tri-band APs, when the dedicated 6 GHz radio is experiencing an outage or recovery, AFC requests may need to be originated from the 2.4 GHz radio to ensure the 6 GHz radio recovers with a valid, fresh AFC authorisation.
- **Band-5 AFC Overlap:** Some 5 GHz radios in APs capable of operating in the 5.925–5.965 GHz sub-band (UNII-5) also require AFC coordination, necessitating AFC origination from the 5 GHz radio interface.

Without radio-aware AFC origination logic, AFC requests are either not submitted for auto-band APs (leaving them unable to operate as standard-power 6 GHz devices) or submitted from incorrect radio interfaces (causing certification ID mismatches and AFC system rejections).

2.5 Deficiency P4: GPS and Location Management Gaps

AFC authorisation is inherently location-dependent: the AFC system grants channel and power permissions based on the device's geographic coordinates and uncertainty ellipse. When an AP moves to a new location—due to physical relocation, building reconfiguration, or GPS sensor drift—the previously issued AFC authorisation may no longer be valid for the new location, as incumbent users at the new location may require protection at different channels or power levels.

Prior AFC implementations use naive geographic comparison (absolute delta in latitude and longitude degrees) to detect location changes. This approach has two critical flaws: (1) a delta of 0.001 degrees in latitude represents approximately 111 metres at the equator but only 55 metres at 60° latitude, making the threshold geographically inconsistent; and (2) altitude changes (e.g., AP relocation between building floors) are not captured in 2D coordinate comparison, potentially missing significant 3D displacement that changes AFC interference geometry.

Additionally, prior implementations do not handle the GPS stabilisation period: immediately after an AP is powered on or claimed, the GPS receiver may report inaccurate coordinates for several minutes while acquiring satellite fixes. Submitting AFC requests during this unstable period results in authorisations for incorrect locations, requiring unnecessary re-registration once GPS stabilises.

Furthermore, when GPS data is unavailable (NoGPS condition), prior implementations immediately disable the 6 GHz radio entirely, eliminating connectivity for all clients on that radio. This aggressive approach is unnecessarily disruptive when the AP model supports Low Power Indoor (LPI)

operation, which does not require AFC authorisation and can maintain reduced-power connectivity while GPS is acquired.

2.6 Deficiency P5: Inadequate EIRP Computation

AFC responses provide two independent power constraint mechanisms: channel-based maxEirp values (one per authorised channel) and frequency-range-based maxPsd values (Power Spectral Density in dBm/MHz across continuous frequency ranges). The binding transmit power limit is the minimum of the EIRP derived from both constraints. Prior implementations either ignore PSD constraints entirely (applying only maxEirp) or apply PSD limits without proper frequency-domain integration across the channel's occupied bandwidth, resulting in either illegal high-power transmissions (when PSD limits are ignored) or unnecessarily conservative power reductions (when PSD limits are applied incorrectly).

The FCC and WINNF-TS-1014 specification mandate minimum power levels per bandwidth (AFC_MIN_PSD): 14 dBm for 20 MHz, 17 dBm for 40 MHz, 20 dBm for 80 MHz, 23 dBm for 160 MHz, and 26 dBm for 320 MHz channels. Prior implementations do not enforce these minimums, potentially configuring APs to operate below regulatory thresholds (wasting spectrum efficiency) or at zero power (effectively disabling the radio).

2.7 Deficiency P6: 320 MHz Wi-Fi 7 Mode Disambiguation

IEEE 802.11be (Wi-Fi 7) introduces 320 MHz channel operation in the 6 GHz band, representing the largest single channel width in consumer Wi-Fi history. Due to the limited 6 GHz spectrum availability (1200 MHz total), only two non-overlapping 320 MHz channel configurations exist:

- Mode 1: Primary Sub-Channel (PSC) at channels 31, 95, and 159; sub-channel range approximately 5945–6265 MHz, 6265–6585 MHz, and 6585–6905 MHz respectively.
- Mode 2: PSC at channels 63, 127, and 191; sub-channel range approximately 5985–6305 MHz, 6305–6625 MHz, and 6625–6945 MHz respectively.

These two modes overlap significantly in frequency: a Mode 1 authorisation at channel 31 partially overlaps with a Mode 2 authorisation at channel 63. Prior AFC implementations do not distinguish between Mode 1 and Mode 2, leading to incorrect EIRP assignment where a Mode 1 PSC EIRP value is applied to Mode 2 sub-channels (or vice versa), violating per-channel power limits. The present invention introduces explicit mode detection via `get_mode(channel)` and independent EIRP expansion for each mode's sub-channel set.

2.8 Deficiency P7: Fragile Claim/Unclaim and Lifecycle Workflows

Enterprise AP deployments involve frequent lifecycle events: APs are provisioned, claimed to sites, assigned to organisations, physically relocated, temporarily disconnected, and eventually decommissioned. Each of these events has specific implications for AFC registration state at the AFC proxy:

- AP Unclaim: When an AP is removed from a Mist organisation (AP_UNCLAIMED), any existing AFC registration at the proxy must be deleted to free the slot for future re-registration under a

potentially different organisation or certification credential.

- AP Unassignment: When an AP is unassigned from a Mist site (AP_UNASSIGNED), the site-specific AFC parameters (org_id, site_id) become invalid, requiring proxy cleanup.
- AP Disconnection: When an AP temporarily loses connectivity, AFC cache in Redis should be preserved (to enable quick recovery) but the GPS data may become stale.

Prior implementations do not propagate these lifecycle events to the AFC proxy, resulting in stale registrations that occupy proxy resources and may block re-registration when the AP is re-claimed (as the proxy may reject duplicate requestId entries). Furthermore, the GPS location cache is not cleared on unclaim, causing the next claim cycle to begin with potentially outdated location data from the AP's previous deployment site.

FIGURES — PART 1

Figure 1

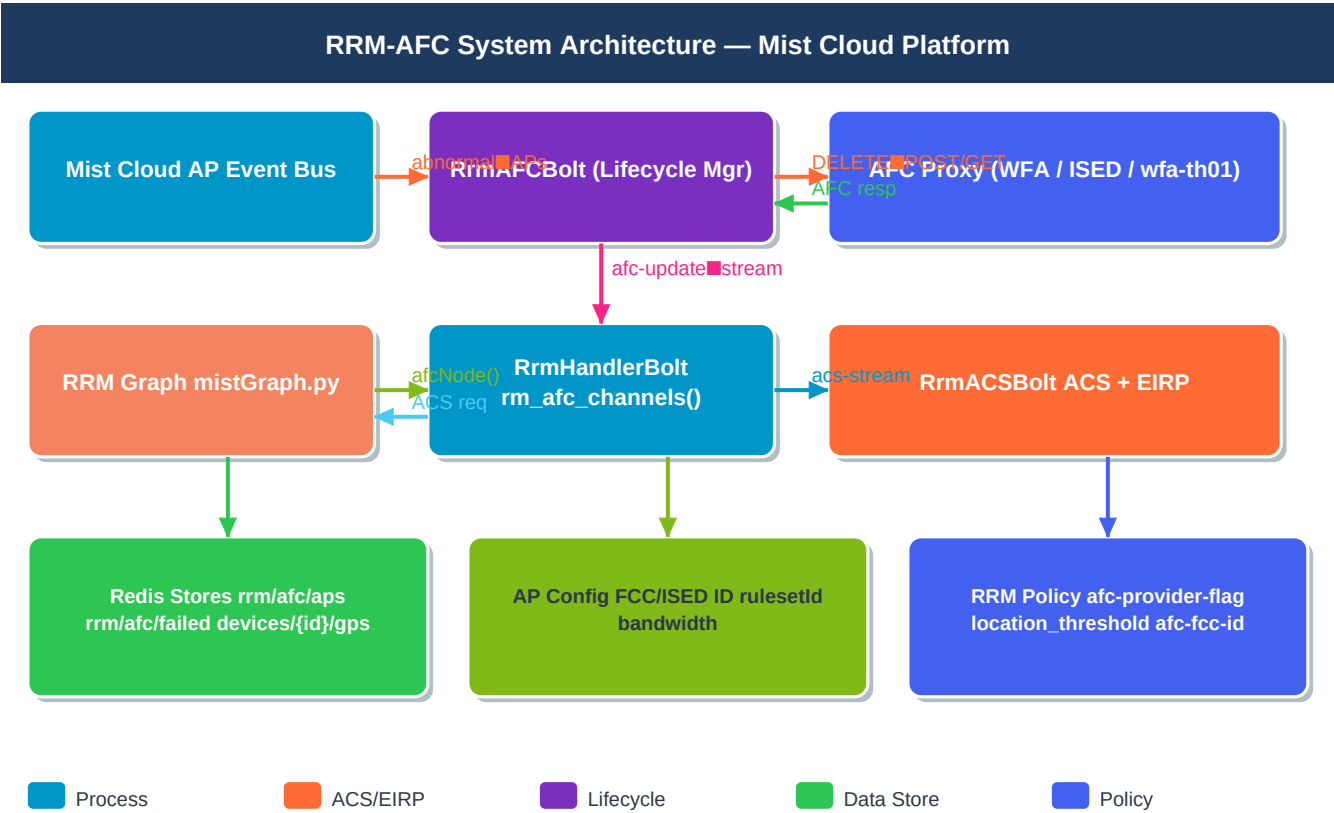


Fig. 1 — End-to-End RRM-AFC System Architecture on Mist Cloud Platform

Figure 2

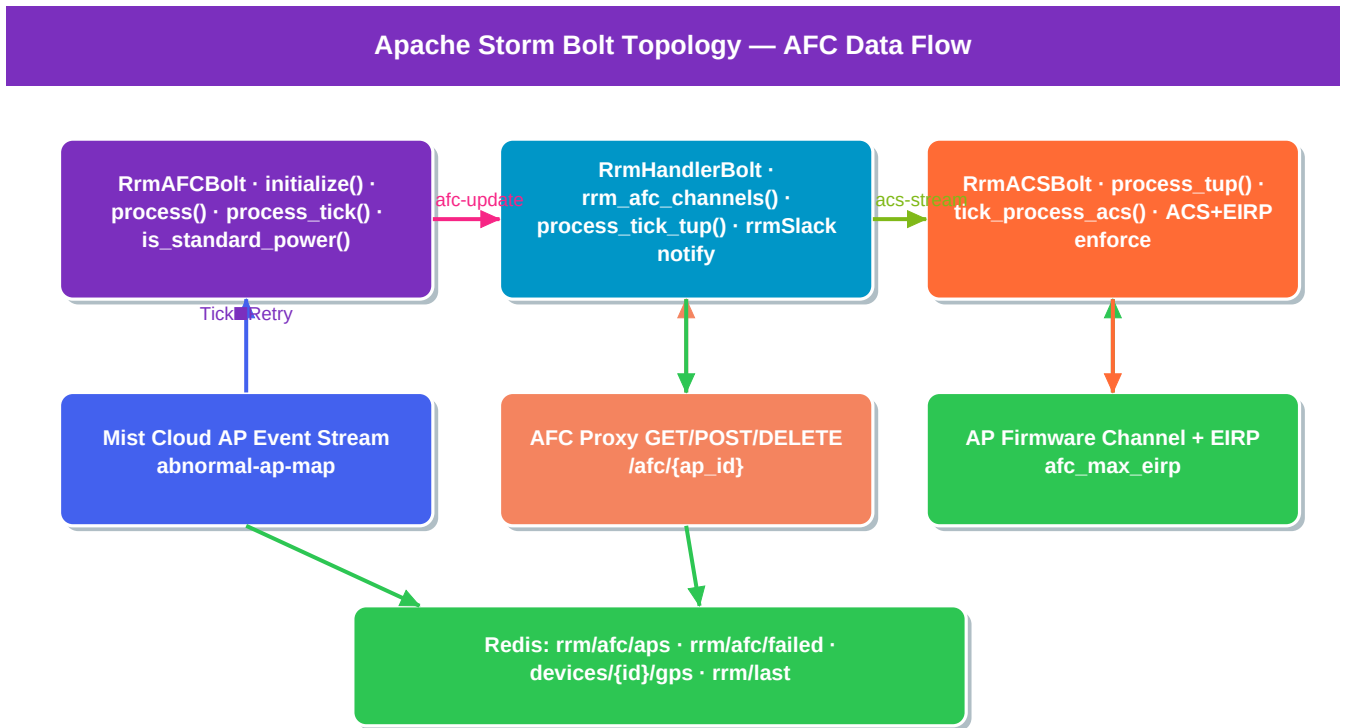


Fig. 2 — Apache Storm Bolt Topology Showing AFC Data Streams

3. SUMMARY OF THE INVENTION

The present invention introduces a comprehensive, RRM-orchestrated AFC control framework implemented as a distributed Apache Storm bolt topology that governs the complete AFC lifecycle—from initial AP registration through response validation, EIRP enforcement, cache management, and AP lifecycle events—on a per-AP, per-band, and per-radio basis. The framework is uniquely distinguished from prior art by the following key innovations:

Innovation 1 — RRM-Driven AFC Request Construction:

The RRM control plane, rather than the AP firmware, constructs and submits AFC request payloads. RRM determines the correct certification identifier (FCC ID for US, ISED ID for Canada), ruleset identifier (resolved dynamically per country code), AP serial number, GPS-derived location ellipse parameters, and the complete set of inquired frequency ranges and global operating classes. This centralised construction ensures consistency across AP models and enables runtime updates without firmware changes.

Innovation 2 — GET-before-POST Query Optimisation:

Before submitting a new AFC POST request, RRM first issues a GET query to the AFC proxy to check for an existing valid entry. Only on cache miss, RETRY state, NULL channel information, or detected location change does RRM issue a fresh POST. This GET-before-POST protocol reduces AFC provider load by up to 80% in stable deployments where AP locations do not change frequently.

Innovation 3 — ECEF-Based Location Change Detection:

AP location changes are detected using Earth-Centered Earth-Fixed (ECEF) coordinate transformation (`lla_to_ecef()`) with WGS84 ellipsoid parameters $a=6,378,137.0\text{m}$, $e^2=6.694\times 10^{-3}$. The Euclidean distance in ECEF space is compared against a configurable threshold (default 200 metres), providing geographically consistent detection independent of latitude. Altitude changes are independently validated via `height_diff` comparison.

Innovation 4 — PSD-to-EIRP Dual-Source Computation:

RRM computes per-channel EIRP as $\min(\max\text{Eirp}[j], \text{psdEirp}[\text{ch}])$ where `psdEirp` is derived by integrating `maxPsd` values from `availableFrequencyInfo` across the channel's occupied frequency range. The result is floored by `AFC_MIN_PSD[bandwidth]` to prevent sub-threshold power assignment. This dual-source computation is expanded to all sub-channels using bandwidth-specific delta arrays.

Innovation 5 — 320 MHz Mode 1/Mode 2 Disambiguation:

Wi-Fi 7 320 MHz channels are explicitly disambiguated into Mode 1 ($\text{PSC} \in \{31,95,159\}$) and Mode 2 ($\text{PSC} \in \{63,127,191\}$) using the `get_mode(channel)` function. EIRP values are expanded only to sub-channels within the correct mode, preventing cross-mode EIRP contamination.

Innovation 6 — Radio-Aware AFC Origination:

RRM selects the correct originating radio for AFC requests based on AP topology: dedicated `r6` for tri-band APs; `auto_band r24` (2.4 GHz radio) for dual-band 2.4+6 GHz APs; and `r5` for APs with 5 GHz UNII-5 overlap. The 2.4 GHz-originated AFC requests include the `afc_standard_power=True` flag to ensure proper standard-power enforcement after band conversion.

Innovation 7 — Dynamic Multi-Provider Selection:

AFC provider selection is driven by per-site RRM policy via the `afc-provider-flag` parameter, mapping to: 'wfa' (production), 'wfa-th01' (internal test harness), 'wfa-th02' (certification harness), or 'dev' (development mock). This enables per-site AFC testing, certification, and production deployment without code changes.

Innovation 8 — NoGPS Resilience with LPI Fallback:

When GPS is unavailable, RRM records the AP's NoGPS condition in Redis with retry metadata (`org_id`, `site_id`, `bandwidth`, `retry_counts`) and, if the AP model supports LPI operation, emits a non-standard-power ACS event to maintain client connectivity. A periodic tick process retries AFC validation for all NoGPS APs, with Slack alerting after 10 consecutive failures.

Innovation 9 — 3-Tier AFC Cache Expiry Policy:

AFC responses are managed under a 3-tier cache policy: Tier 1 (time-to-expiry > 2h) reuses cached channels without any AFC query; Tier 2 (0-2h) issues a GET validation; Tier 3 (expired or location-changed) performs full DELETE-POST-GET cycle. A maximum 16-hour expiry interval is enforced independently of the AFC provider's stated availability time.

Innovation 10 — Atomic AP Lifecycle Workflows:

AP_UNCLAIMED and AP_UNASSIGNED events trigger a coordinated three-step atomic cleanup: (1) DeleteAP() with retry logic at the AFC proxy; (2) `del_afc_info()` clearing both `rrm/afc/aps` and `rrm/afc/failed` Redis hashes; (3) GPS cache deletion from `devices/{apID}/gps`. This ensures clean re-registration after AP re-claim.

Innovation 11 — AP Outage Recovery Re-Authorisation:

AP radio outage recovery events in standard-power mode trigger a dedicated `ap-outage-recovery-afc` path that performs fresh AFC re-validation from the 2.4 GHz radio, ensuring the recovering 6 GHz radio is configured with a current, valid AFC authorisation rather than a potentially expired cached response.

4. DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

The following detailed description sets forth specific embodiments of the invention with reference to the accompanying figures. The embodiments described herein are implemented in the `mistsys/mist-rrm-exp` codebase as part of the Mist Cloud RRM platform. Unless otherwise specified, all function names, class names, and variable names correspond to identifiers in the reference implementation.

4.1 System Architecture Overview

Figure 3

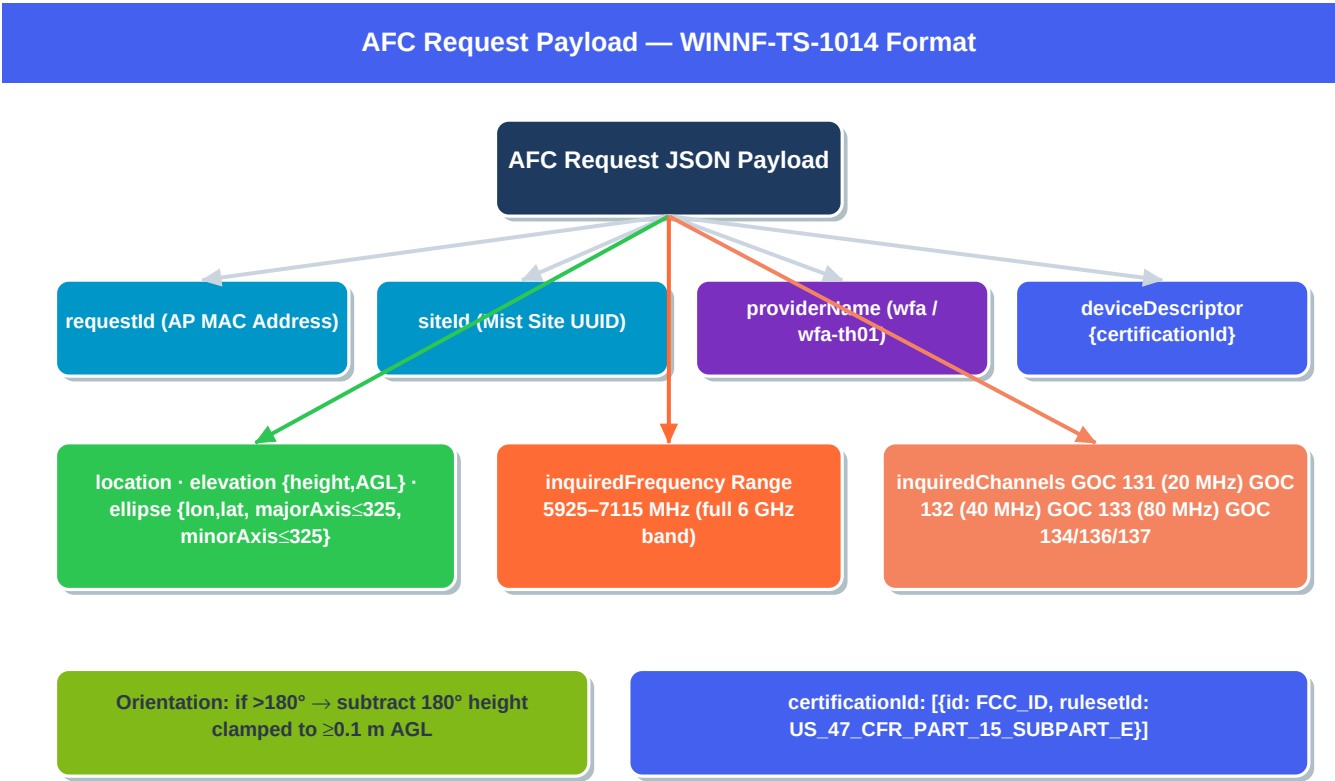


Fig. 3 — AFC Request Payload Construction (WINNF-TS-1014)

Referring to Figure 1 (Part 1), the invention is implemented as a three-bolt Apache Storm topology operating within the Mist Cloud RRM platform. The three primary processing components are: (1) `RrmAFCBolt`, the AP lifecycle manager responsible for claim/unclaim events and periodic retry; (2) `RrmHandlerBolt`, the AFC response router that receives authorisation updates from the AFC proxy and injects them into the ACS pipeline; and (3) `RrmACSBolt`, the channel selection engine that integrates AFC channel constraints and EIRP limits into all channel assignment decisions.

The `RrmAFCBolt` (implemented in `rrmAFC.py`) initialises with connections to two independent Redis pools: the RRM Redis (`rrm-proxy.location`) for AFC state management, and the Location Redis

(redis.location.location) for GPS data retrieval. The bolt maintains an in-memory `ap_standard_power_cache` that records which APs are standard-power capable, enabling efficient filtering of AP lifecycle events. On each incoming tuple from the Mist Cloud AP event stream, the bolt extracts the abnormal AP map (a dictionary of `{ap_id → {ap_status, site_id, org_id}}`) and filters for APs with `AP_UNCLAIMED` or `AP_UNASSIGNED` status.

The `RrmHandlerBolt` (`rrmHandler.py`, `rrm_afc_channels()` method) receives AFC authorisation updates from the `afc-proxy` response stream. Upon receiving a tuple with `state='DONE'` and non-null `availableChannelInfo`, the bolt constructs an RRM command with `command='afc-update'` and emits it to the `acs-stream` for processing by `RrmACSBolt`. The handler validates that the AFC response is complete before forwarding, discarding incomplete or error responses with a `logWarn` entry.

Three Redis data stores underpin the system: (1) `rrm/afc/aps` (hash): active AFC retry queue containing APs in `NoGPS`, `NotDONE`, or `RETRY` state; (2) `rrm/afc/failed` (hash): permanently failed APs with `retry_counts ≥ 10`; and (3) `devices/{apID}/gps` (string key): GPS location data from the Location Redis service.

4.2 Apache Storm Bolt Topology

Figure 4

afc_channel_validation() — GET-before-POST Flow



Fig. 4 — afc_channel_validation() GET-before-POST Decision Flow

The RrmACSBolt (rrmACS.py and rrmACSV2.py) is the central integration point between AFC and ACS. When an ACS request arrives with `afc_standard_power=True`, the bolt first checks for pre-validated `afc_channels` in the command tuple. If `afc_channels` are absent (e.g., for a new AP not yet in the AFC cache), the bolt invokes `afc_channel_validation()` inline to obtain current authorisation. The bolt then calls `channels_validation()` to intersect the AFC-authorized channel set with the AP's configured channel list, ensuring only channels present in both sets are considered for ACS.

After ACS selects an optimal channel based on interference metrics, the bolt assigns `afc_max_eirp` from the validated channel map. If the selected channel is not found in the map (a defensive condition), the bolt selects a random channel from the AFC-authorized set using `tools.random_choice()` and assigns its corresponding EIRP. This ensures the AP always operates on an authorised channel with a valid power limit.

4.3 AFC Request Payload Construction

The AFC request payload is constructed by `get_payload_proxy()` in `afc_payload.py`, conformant to WINNF-TS-1014 AFC System to AFC Device Interface Specification version 1.4. The payload JSON structure contains the following mandatory and optional fields:

- `requestId`: Set to the AP's MAC address (lower-case, colon-separated). Used as the unique identifier for GET and DELETE operations at the AFC proxy.
- `siteId`: The Mist site UUID associated with the AP's current assignment.
- `providerName`: Dynamically resolved from the `afc-provider-flag` RRM policy parameter via `get_flag_provider_mapping()`. Defaults to 'wfa' (Wi-Fi Alliance production system).
- `deviceDescriptor.serialNumber`: Set to the AP MAC address.
- `deviceDescriptor.certificationId`: An array containing a single object `{id, rulesetId}`. For US deployments, `id` is the FCC ID (e.g., '2AHBN-AP64') and `rulesetId` is 'US_47_CFR_PART_15_SUBPART_E'. For Canadian deployments, `id` is the ISED ID and `rulesetId` is resolved via `get_rulesetId_from_country_code('CA')`.
- `location.elevation`: Contains height (AGL metres, clamped to ≥ 0.1 m), `heightType` ('AGL' — Above Ground Level), and `verticalUncertainty` from GPS data.
- `location.ellipse`: Contains center `{longitude, latitude}`, `majorAxis` (capped to 325m), `minorAxis` (capped to 325m), and `orientation` (normalised: if $> 180^\circ$, subtract 180°).
- `location.indoorDeployment`: Set to 1 (conservative indoor assumption).
- `inquiredFrequencyRange`: `[{lowFrequency: 5925, highFrequency: 7115}]` covering the full 6 GHz band as allocated for Wi-Fi use.
- `inquiredChannels`: Array of `{globalOperatingClass}` objects covering GOC 131 (20 MHz), 132 (40 MHz), 133 (80 MHz), 134 (160 MHz), 136 (80+80 MHz / non-UNII 5/7), and 137 (320 MHz).

The orientation normalisation (subtract 180° if $> 180^\circ$) is a critical detail: GPS ellipse orientation is defined as the angle of the major axis from true north, in the range $[0^\circ, 360^\circ)$. The AFC system expects orientation in $[0^\circ, 180^\circ)$, so orientations in the upper semicircle are mapped to their equivalent lower-semicircle representation. This mapping preserves the geometric meaning of the uncertainty ellipse while satisfying the AFC system's input validation constraints.

4.4 GET-before-POST Request Lifecycle

The `afc_channel_validation()` function in `afc_utils.py` implements the core AFC query logic. Refer to Figure 4 for the complete decision flow. The function accepts the following parameters: `afc_proxy_link` (URL), `site_id`, `ap_id`, `bandwidth` (default 80), `redis_conn`, `location_redis_conn`, `provider_flag` (default '0'), `afc_response` (cached response for reuse), `afc_certificationId`, and `location_threshold` (default 200m).

The function first checks if `provider_flag` is 'dev'. In development mode, it attempts to retrieve a mock AFC response from Redis using `get_mock_afc_response()`. If no mock response is stored, it falls back to `get_default_response(ap_id, site_id)`, which returns a synthesised DONE response with representative channels and EIRPs for testing purposes. This mode enables end-to-end AFC workflow validation without querying the production WFA system.

For non-development mode, the function retrieves the AP's current GPS coordinates via `get_ap_geolocation(location_redis_conn, ap_id)`. If GPS data is unavailable, the function returns `{}`,

None, 'NoGPS'), triggering the NoGPS handling path described in Section 4.10. If GPS is available, it constructs the AFC payload using `get_payload_proxy()` and issues a `GET /afc/{ap_id}` request via `GetChannelByMac()`. The `GET` response is checked for: (1) `state == 'DONE'`; (2) `availableChannelInfo` not None; and (3) `location` unchanged (via `location_match()`). If all three conditions are met, the response is passed to `process_afc_response()` without a new `POST`.

On any failure condition (`state != DONE`, null channels, or location change), the function issues `PostAfcRequest()` followed by a second `GetChannelByMac()` to retrieve the provider's response. The state from the second `GET` is checked: 'DONE' with non-null `availableChannelInfo` triggers `process_afc_response()`; any other state returns (`{}`, None, 'NotDONE'), causing the ACS bolt to defer channel assignment.

4.5 GPS Geolocation Management

Figure 5

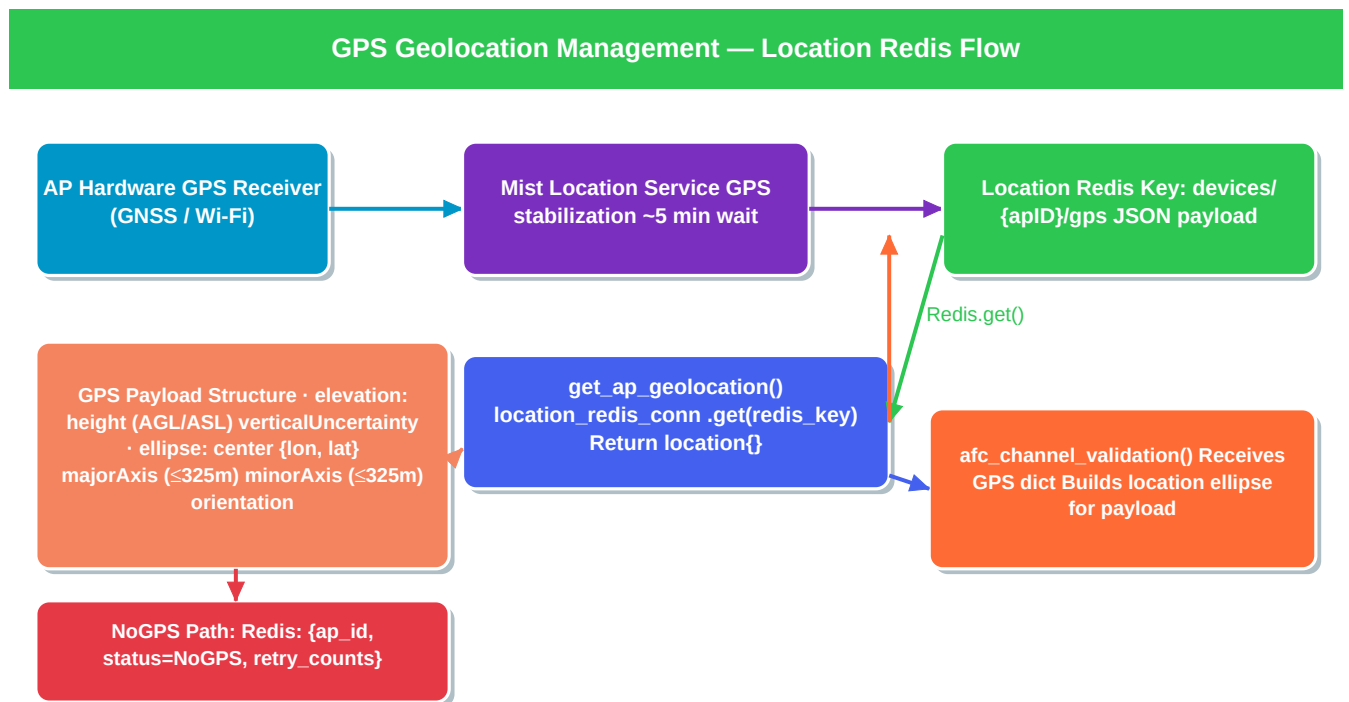


Fig. 5 — GPS Geolocation Data Flow and `get_ap_geolocation()`

GPS data is stored in the Location Redis service under the key 'devices/{apID}/gps' as a JSON string. The `get_ap_geolocation()` function retrieves this data and returns the nested 'location' dictionary, which contains 'elevation' (with height in metres, heightType as 'AGL' or 'ASL', and verticalUncertainty) and 'ellipse' (with center {longitude, latitude}, majorAxis in metres, minorAxis in metres, and orientation in degrees).

The GPS data is populated by the Mist Location Service, which aggregates positioning data from: (1) the AP's onboard GPS receiver (where present); (2) Wi-Fi-based positioning using multilateration against known reference points; and (3) manual coordinate entry via the Mist dashboard. For APs

with onboard GPS receivers, a stabilisation period of approximately 5 minutes is observed after initial power-on or re-claim before GPS coordinates are considered reliable. The tick process in RrmAFCBolt handles this stabilisation by recording the AP in the NoGPS retry queue and retrying AFC validation on each subsequent tick until stable GPS is available.

The GPS data structure includes a timestamp in Unix microseconds, enabling the system to detect GPS data staleness independently of the AFC response expiry. When an AP is unclaimed, the GPS cache entry is explicitly deleted from Location Redis (location_redis_conn.delete('devices/{apID}/gps')), ensuring the next claim cycle does not inherit stale location data from the AP's previous deployment site.

Figure 6

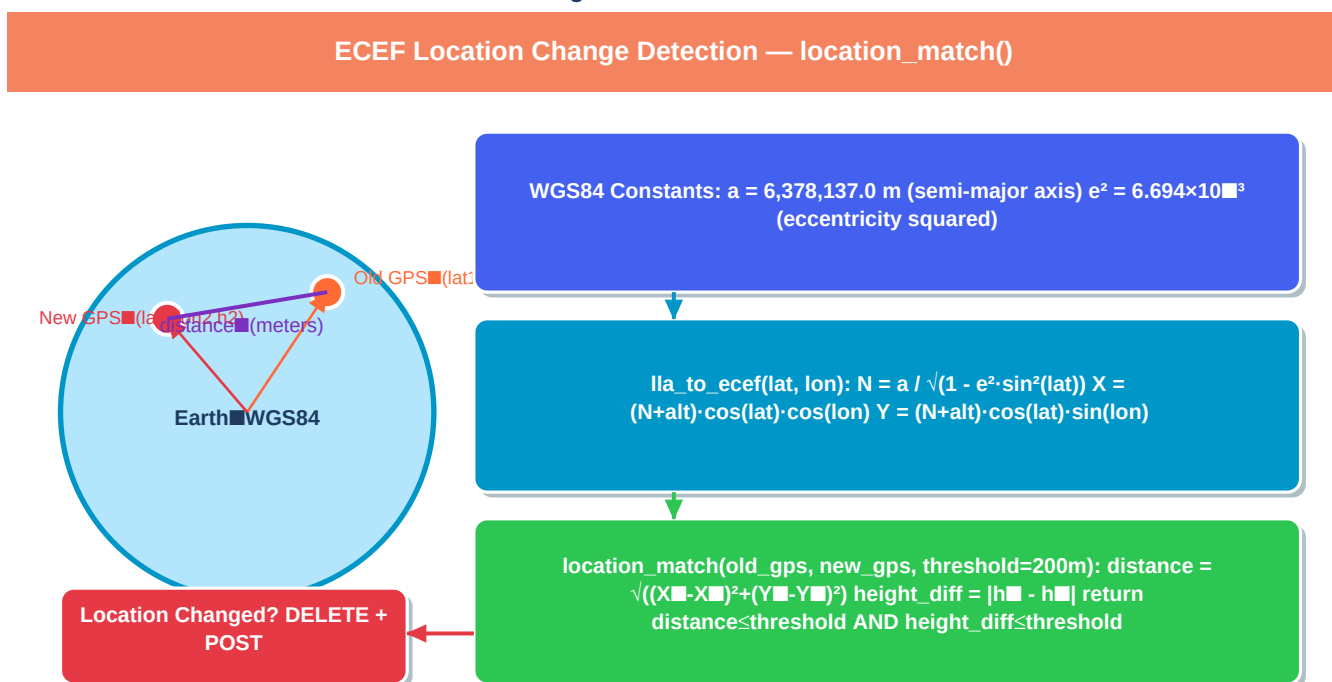


Fig. 6 — ECEF-Based Location Change Detection (location_match())

Figure 7

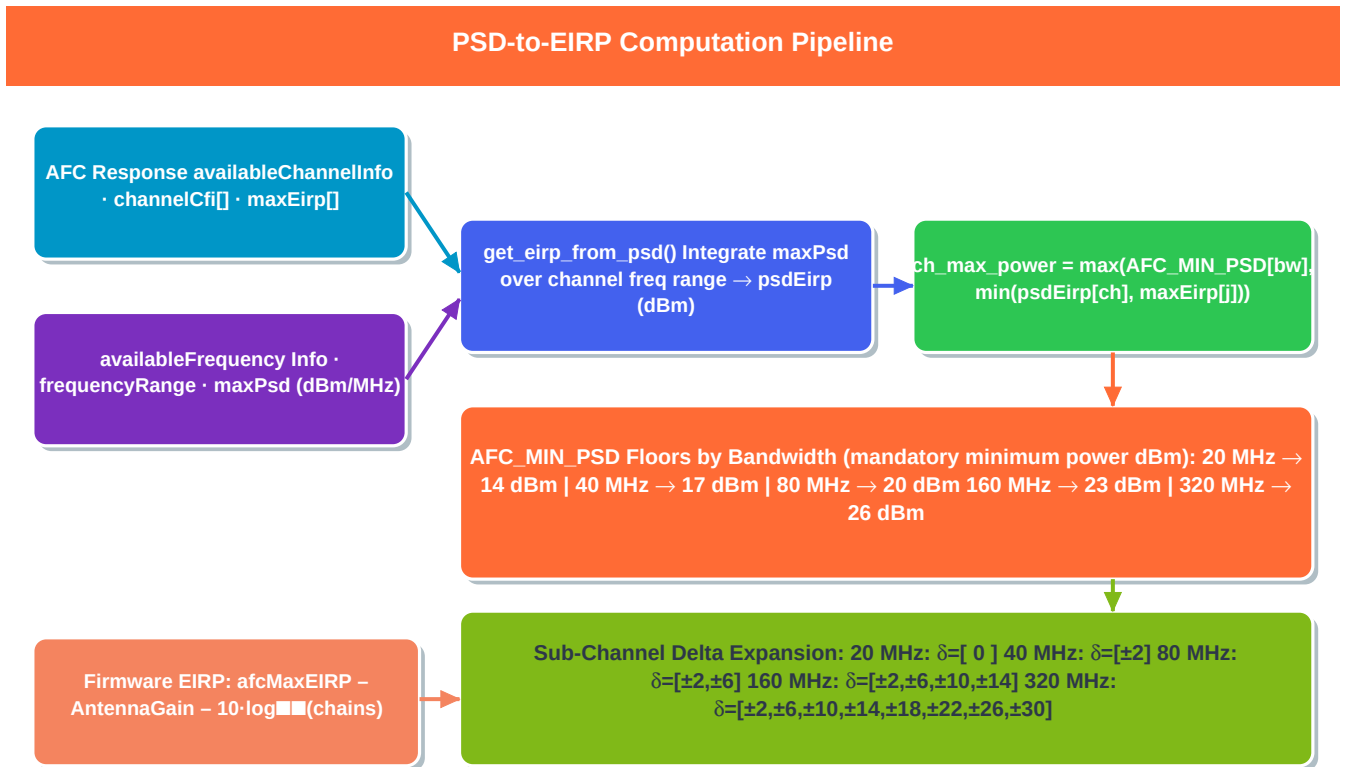
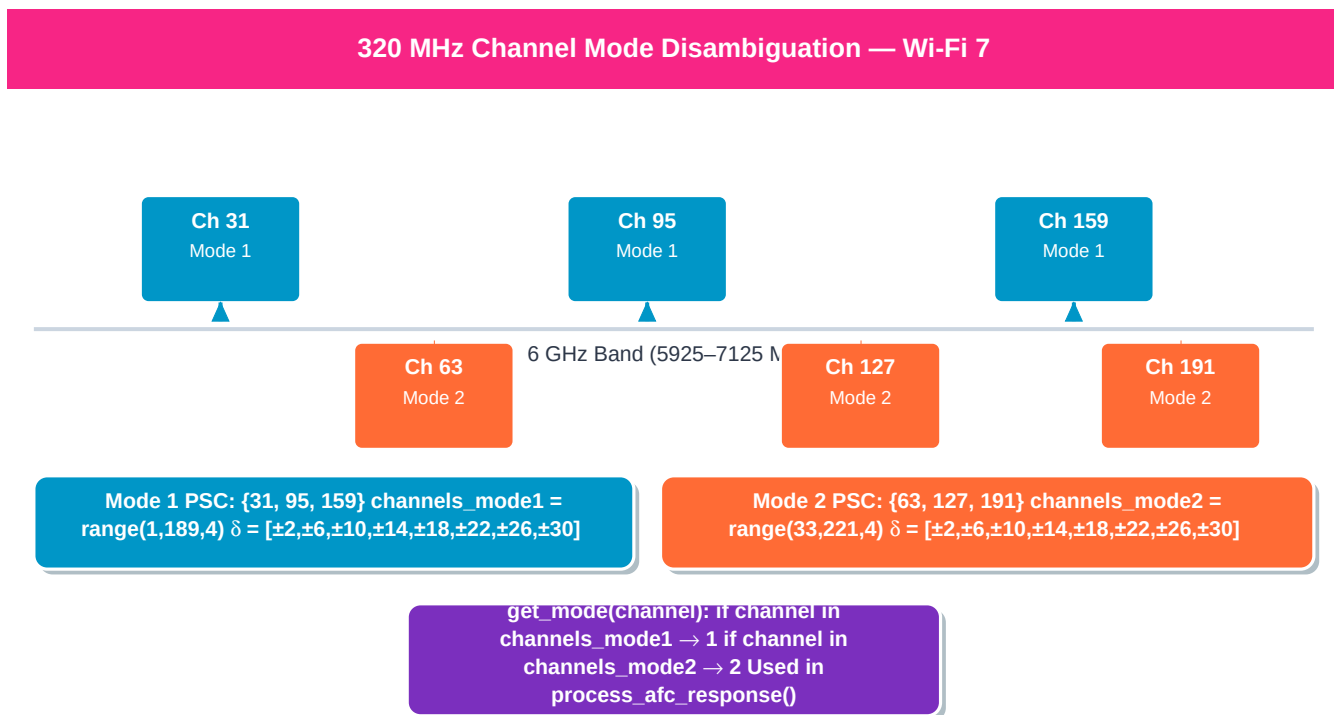


Fig. 7 — PSD-to-EIRP Computation and AFC_MIN_PSD Floor Enforcement

4. DETAILED DESCRIPTION (continued)

4.6 ECEF-Based Location Change Detection

Figure 8



* Mode 1 and Mode 2 have distinct non-overlapping PSC sets. EIRP must be assigned only to sub-channels within the correct 320 MHz mode.

Fig. 8 — 320 MHz Wi-Fi 7 Channel Mode 1/Mode 2 Disambiguation

Accurate location change detection is critical to AFC compliance: when an AP moves beyond the geographic extent of its authorised location ellipse, previously issued channel and power permissions may no longer be valid for the new location. The location_match() function implements ECEF-based distance computation to provide geographically consistent displacement detection.

The WGS84 Earth ellipsoid is defined by semi-major axis $a = 6,378,137.0$ metres and first eccentricity squared $e^2 = 6.6943799901414 \times 10^{-3}$. The lla_to_ecef() function converts geodetic coordinates (latitude ϕ , longitude λ , altitude h) to ECEF Cartesian coordinates (X , Y , Z) using the standard transformation:

$$\begin{aligned} N(\phi) &= a / \sqrt{1 - e^2 \cdot \sin^2(\phi)} && [\text{radius of curvature in prime vertical}] \\ X &= (N + h) \cdot \cos(\phi) \cdot \cos(\lambda) \\ Y &= (N + h) \cdot \cos(\phi) \cdot \sin(\lambda) \\ Z &= (N \cdot (1 - e^2) + h) \cdot \sin(\phi) \end{aligned}$$

The Euclidean distance between two ECEF positions is computed as $\sqrt{(X_{\text{new}} - X_{\text{old}})^2 + (Y_{\text{new}} - Y_{\text{old}})^2}$, representing the horizontal displacement in metres. Height displacement is independently computed as $|h_{\text{new}} - h_{\text{old}}|$. The AP is considered to have changed location if either displacement exceeds the

configured threshold (location_threshold, default 200 metres per RRM policy). The independent height check captures floor changes in multi-storey buildings that would not be reflected in the horizontal distance calculation.

The location_threshold parameter is configurable per Mist site via the RRM policy system (rrm_policy.get_rrm_policy()), enabling operators to set tighter thresholds for high-precision indoor deployments (e.g., 50 metres in dense enterprise environments) or looser thresholds for outdoor deployments where GPS uncertainty is higher (e.g., 500 metres). The test test_afc_tick_process_location_threshold validates this configurable behaviour with a threshold of 50,000 metres.

When location change is detected, RRM performs an atomic operation: (1) DeleteAP(afc_proxy_link, res.get('requestId')) removes the stale entry from the AFC proxy; (2) the new AFC payload is constructed with the current GPS coordinates; (3) PostAfcRequest() submits the fresh registration; and (4) GetChannelByMac() retrieves the new authorisation. This atomic sequence ensures no gap in compliance: the AP remains offline (or in LPI mode) until the new authorisation is obtained.

4.7 PSD-to-EIRP Conversion and EIRP Enforcement

AFC responses provide two complementary power constraint mechanisms that must both be satisfied. The process_afc_response() function implements dual-source EIRP computation via the following algorithm:

```
globalOperatingClass = bandwidth_to_globalOperatingClass(bandwidth)
# Maps: 20→131, 40→132, 80→133, 160→134, 320→137

For each channelClass in availableChannelInfo
  where globalOperatingClass matches:

    For each channel ch in channelClass.channelCfi[]:
      psdEirp[ch] = get_eirp_from_psd(channel_freqs, availableFreqInfo, bw)
      ch_max_power = max(AFC_MIN_PSD[bw],
                        min(psdEirp[ch], channelClass.maxEirp[j]))

    # Expand to sub-channels based on bandwidth:
    for delta in ch_delta[bandwidth]:
      if (ch ± delta) in channels_6G_20:
        afc_channels[ch ± delta] = ch_max_power
```

The get_eirp_from_psd() function integrates the maxPsd values over the channel's occupied frequency range. For each 20 MHz sub-channel within the occupied bandwidth, the function finds the PSD entry in availableFrequencyInfo whose frequencyRange encompasses the sub-channel's centre frequency and applies the corresponding maxPsd value. The EIRP equivalent is computed as $\text{maxPsd} + 10 \cdot \log_{10}(\text{channel_bandwidth_MHz})$, converting the spectral density limit to a total-power EIRP limit.

The AFC_MIN_PSD floors are mandatory regulatory minimums specified in FCC KDB 987594: 20 MHz → 14 dBm, 40 MHz → 17 dBm, 80 MHz → 20 dBm, 160 MHz → 23 dBm, 320 MHz → 26 dBm. These floors ensure that the computed EIRP never falls below the minimum level required for useful operation, preventing the system from assigning sub-threshold power levels that would result in

ineffective transmissions.

The resulting `afc_channels` dictionary (`{channel_number → max_eirp_dBm}`) is passed through `channels_validation()` to intersect with the AP's configured channel list, then stored in the RRM graph node (`g.node[ap_rad_id]['afc_channels']`) for subsequent ACS decisions. The `afc_max_eirp` value assigned to the ACS-selected channel is passed downstream to the AP firmware, which applies: $\text{tx_power} = \text{afc_max_eirp} - \text{antenna_gain} - 10 \cdot \log_{10}(\text{radio_chains})$.

4.8 320 MHz Wi-Fi 7 Channel Mode Disambiguation

Figure 9

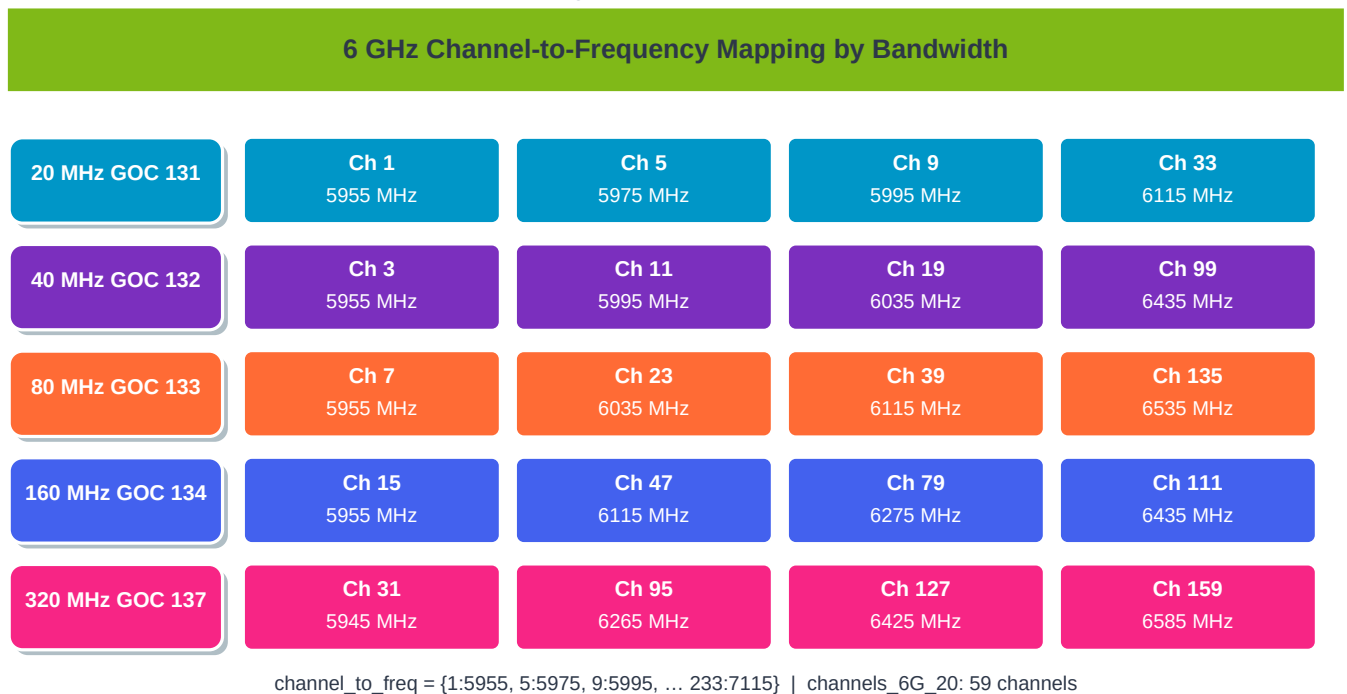


Fig. 9 — 6 GHz Channel-Frequency Mapping per Bandwidth (20–320 MHz)

The 320 MHz channel mode disambiguation addresses a unique challenge in Wi-Fi 7 deployments. The 6 GHz band (1200 MHz of usable spectrum) can accommodate at most three non-overlapping 320 MHz channels in each of two mode configurations. The two modes share sub-channel frequencies but have different primary sub-channel (PSC) assignments, creating a potential for incorrect EIRP assignment if modes are conflated.

The implementation defines two channel sets derived from the respective PSC arrays:

```
central_channel_mode1 = [31, 95, 159]
central_channel_mode2 = [63, 127, 191]

# Sub-channel range around each PSC (step=4, span=±30 channels):
channels_mode1 = range(central_mode1[0]-30, central_mode1[-1]+34, 4)
# → [1, 5, 9, ..., 159, 163, ..., 189]

channels_mode2 = range(central_mode2[0]-30, central_mode2[-1]+34, 4)
# → [33, 37, 41, ..., 191, 195, ..., 221]
```

```
def get_mode(channel):
    if channel in channels_mode1: return 1
    if channel in channels_mode2: return 2
    return 0 # unknown
```

During EIRP expansion in `process_afc_response()`, when `bandwidth == 320`, the function determines the mode of the PSC using `get_mode(ch)` and applies the delta expansion only to sub-channels within the same mode: `afc_channels[sub_ch] = ch_max_power` only when `get_mode(sub_ch) == get_mode(ch)`. This prevents a Mode 1 PSC authorisation from contaminating Mode 2 sub-channels and vice versa, ensuring per-channel EIRP compliance across the 320 MHz occupied bandwidth.

4.9 Channel-to-Frequency Mapping

Figure 10

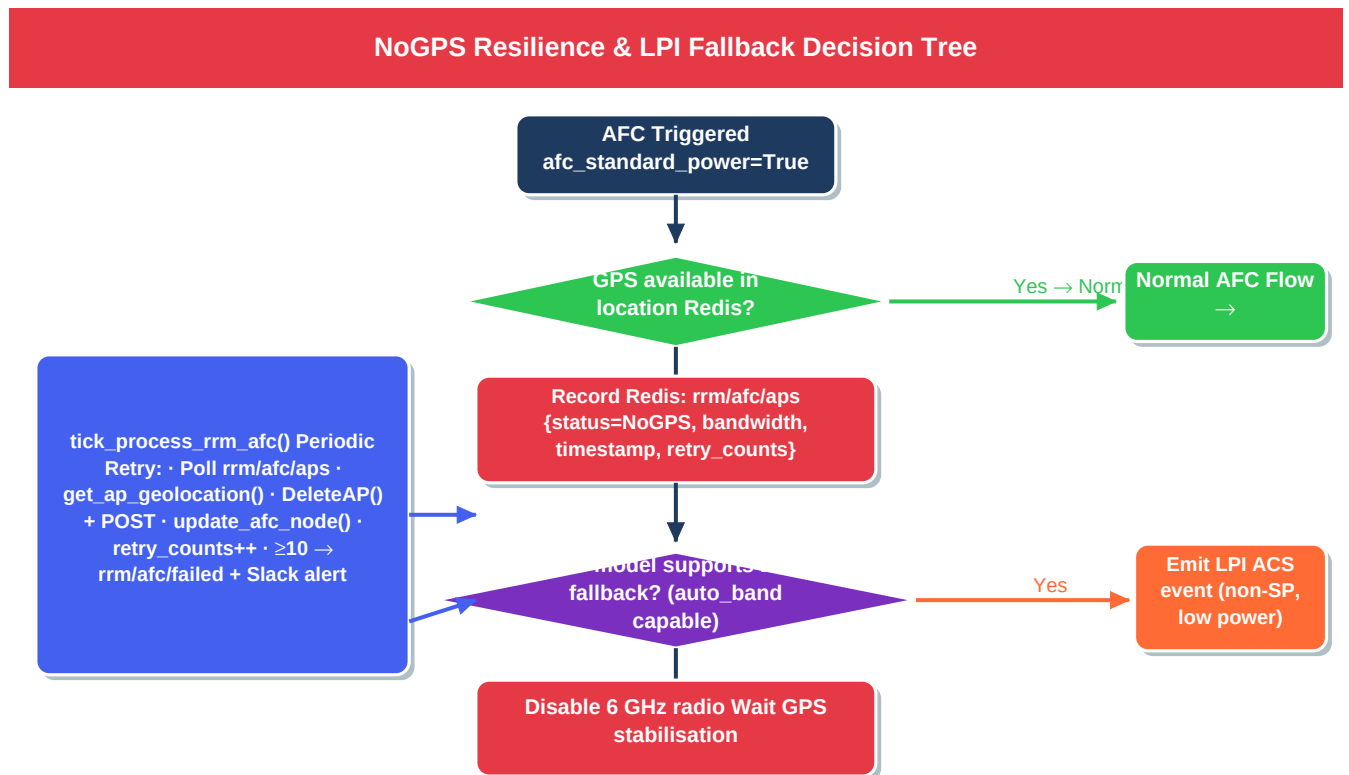


Fig. 10 — NoGPS Resilience and LPI Fallback Decision Tree

The `get_channel_freq_mapping()` function in `afc_payload.py` provides the definitive mapping between 6 GHz channel numbers (CFI — Channel Frequency Index) and their corresponding frequency ranges. This mapping is used in two contexts: (1) payload construction, where channel numbers are converted to frequency ranges for the `inquiredFrequencyRange` field; and (2) EIRP computation, where the channel's frequency range is used to integrate PSD values.

The 6 GHz band supports 59 valid 20 MHz channels (channels_6G_20 list), starting at channel 1 (centre frequency 5955 MHz) and ending at channel 233 (centre frequency 7115 MHz), with a 20 MHz spacing ($\text{channel_to_freq}[n] = 5935 + n \times 5 \text{ MHz}$). The full channel-to-frequency dictionary is defined as: $\text{channel_to_freq} = \{1:5955, 5:5975, 9:5995, \dots, 229:7095, 233:7115\}$.

For wider bandwidths, the channel mapping nests sub-channels within their parent bandwidth structure. For example, the 80 MHz channel at CFI 7 spans 5955–6035 MHz and contains 20 MHz sub-channels at CFI 1, 5, 9, 13. The EIRP expansion algorithm uses the `ch_delta` array to identify all sub-channels of a given PSC, ensuring consistent EIRP assignment across the full occupied bandwidth.

4.10 NoGPS Resilience and LPI Fallback

The NoGPS condition arises when `afc_channel_validation()` cannot retrieve valid GPS coordinates for an AP from the Location Redis. This can occur during: (1) initial AP provisioning before GPS fix acquisition; (2) temporary GPS service interruption; (3) AP relocation before GPS update propagation; or (4) GPS hardware failure. The system implements a two-tier response to NoGPS:

Tier A — LPI Fallback (for LPI-capable APs):

If the AP model supports Low Power Indoor (LPI) operation—indicated by the combination of `afc_standard_power=True` and `auto_band` capability in the AP config—RRM emits a non-standard-power ACS event with reduced power limits (LPI maximum 24 dBm EIRP). The AP operates in LPI mode, maintaining client connectivity at reduced power without AFC authorisation. This is implemented in the `afcNode()` function (`APnode.py`) and validated by the `test_rrm_global_afc_nogps_lpi_fallback` test case, which verifies that the standard-power flag is `False` and the radio is not disabled.

Tier B — Redis Tracking and Periodic Retry:

Regardless of LPI fallback, the NoGPS condition is always recorded in Redis for audit and retry purposes. The `set_new_afc_node()` function records: `{org_id, site_id, ap_id, ap_status='NoGPS', bandwidth, timestamp, retry_counts=1}`. The `update_afc_node()` function increments `retry_counts` on each subsequent NoGPS occurrence. When `retry_counts` reaches 10, the AP is moved to the `rrm/afc/failed` hash and a Slack alert is sent to `#afc-rrm-alert-test`.

The `tick_process_rrm_afc()` function, called on every Storm tick by `RrmAFCBolt`, polls all entries in `rrm/afc/aps` and attempts GPS retrieval for each. Upon successful GPS acquisition, it executes `DeleteAP()` to clear any stale proxy entry, then `afc_channel_validation()` to obtain fresh authorisation. If successful (`DONE`), `del_afc_info()` removes the AP from the retry queue.

Figure 11

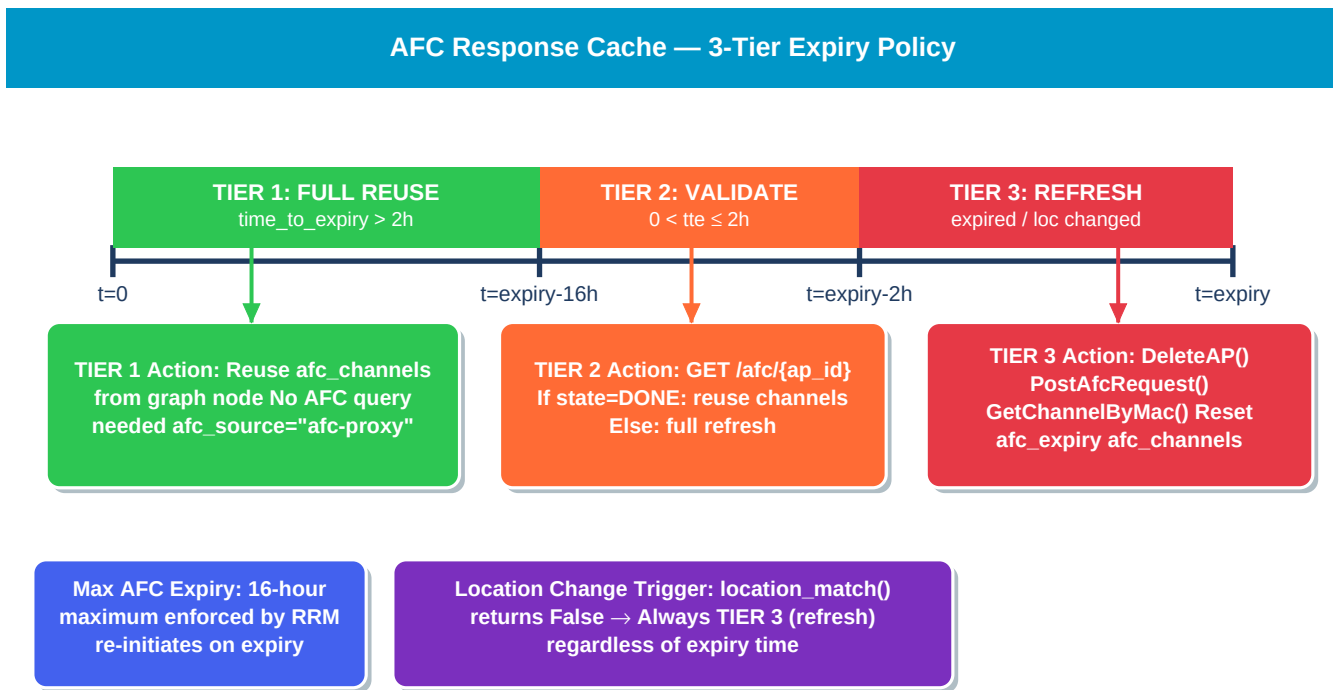


Fig. 11 — 3-Tier AFC Response Cache Expiry and Reuse Policy

Figure 12

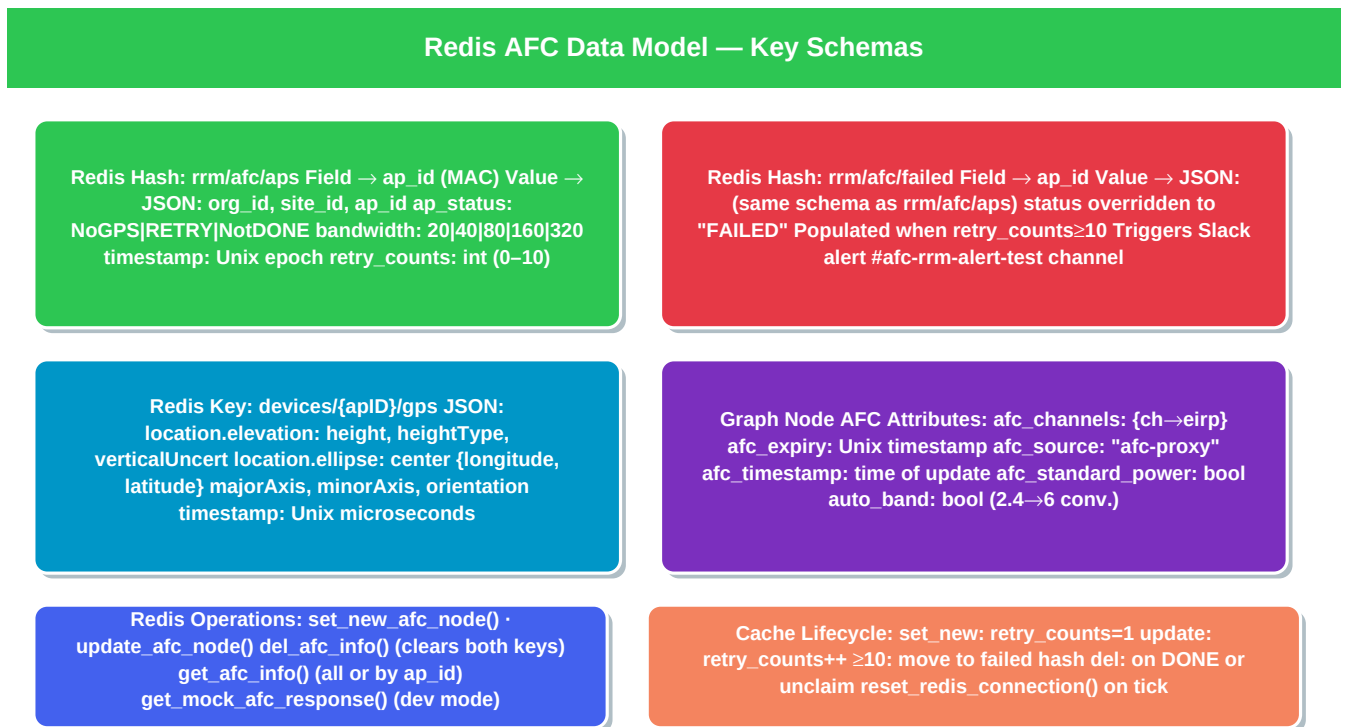


Fig. 12 — Redis AFC Data Model — Key Schemas and Operations

4. DETAILED DESCRIPTION (continued)

4.11 Three-Tier AFC Cache Expiry and Reuse Policy

Figure 13

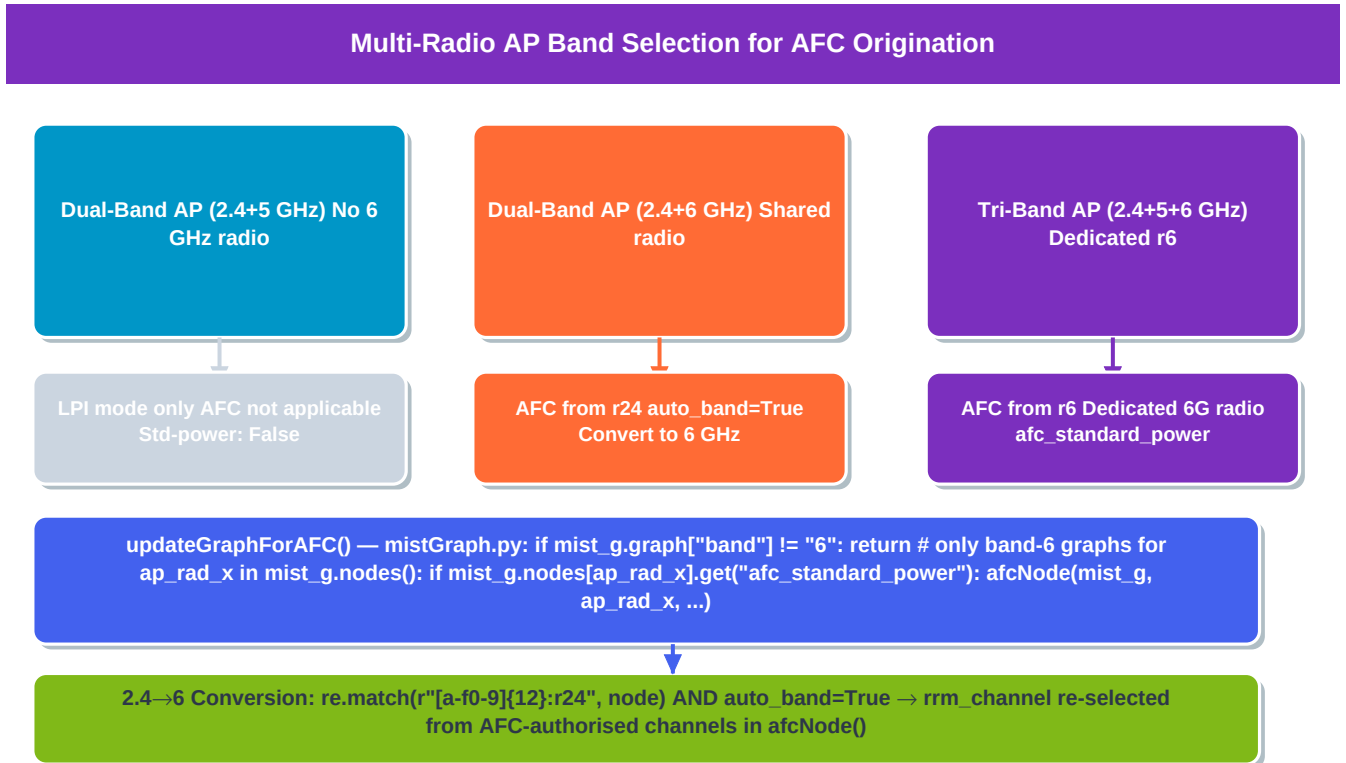


Fig. 13 — Multi-Radio AP Band Selection and updateGraphForAFC()

AFC responses have a finite validity period specified by the `availabilityExpireTime` field in the AFC response. RRM enforces a maximum 16-hour expiry regardless of the provider-stated expiry to ensure regulatory freshness. The cache management is implemented in `afcNode()` with three distinct tiers:

- Tier 1 (`time_to_expiry > 2` hours): The existing `afc_channels` are reused from the graph node without any AFC query. The `afc_source` field records 'afc-proxy' and the `afc_timestamp` records the original update time. This tier handles the majority of RRM cycles in stable deployments.
- Tier 2 (`0 < time_to_expiry ≤ 2` hours): RRM issues a `GET /afc/{ap_id}` to validate the cached entry is still DONE. If the provider returns a fresh DONE response, the cached channels are updated with the new expiry. If not, a full Tier 3 refresh is triggered.
- Tier 3 (`time_to_expiry ≤ 0` or location changed): Full DELETE-POST-GET cycle is executed. The `afc_expiry` in the graph node is reset to the new response's `availabilityExpireTime`, parsed from ISO8601 UTC format and converted to Unix timestamp.

The location change trigger bypasses all tier checks: any detected displacement beyond the configured threshold always triggers Tier 3, regardless of remaining expiry time.

4.12 Redis AFC Data Model

Figure 14

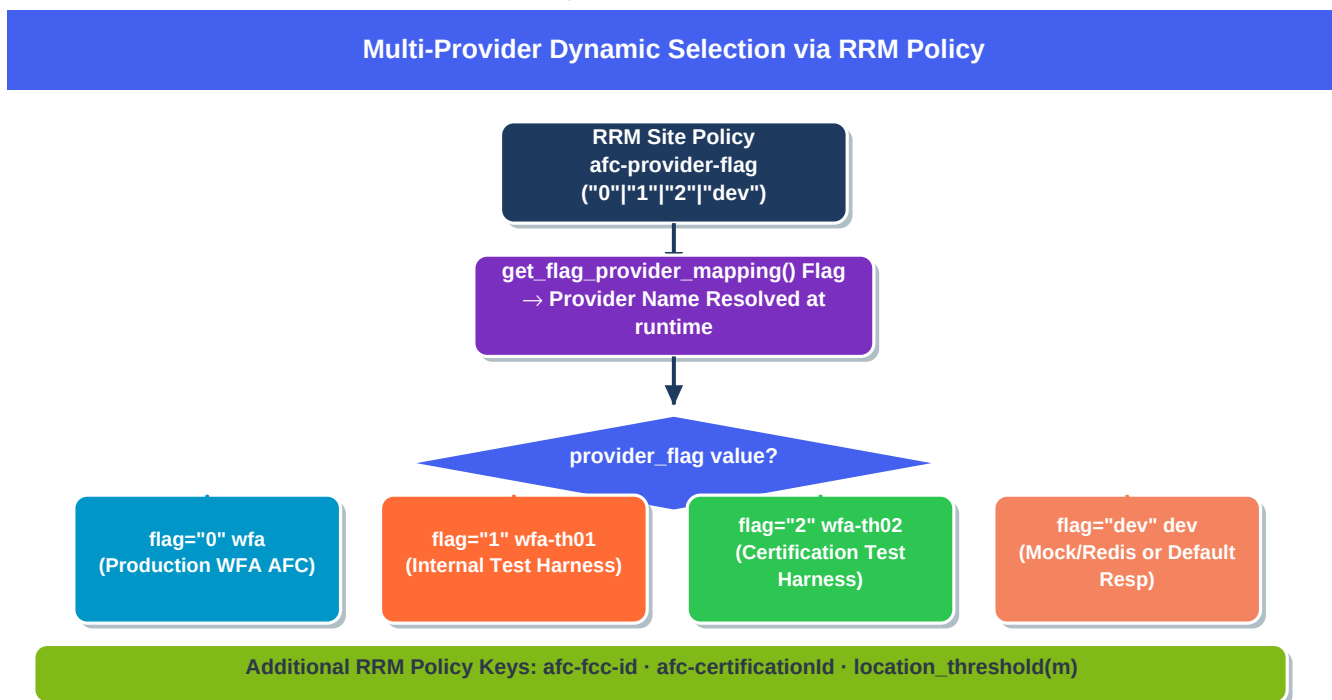


Fig. 14 — Multi-Provider Dynamic Selection via RRM Policy (`get_flag_provider_mapping()`)

The Redis AFC data model consists of two hash keys and one string key per AP. The `set_new_afc_node()` and `update_afc_node()` functions in `rrmRedis.py` manage the `rrm/afc/aps` hash, while `del_afc_info()` provides atomic cleanup of both the active and failed hashes.

The `update_afc_node()` function implements retry count management: it retrieves the existing entry, increments `retry_counts`, and re-saves. When `retry_counts` reaches 10, the AP is moved to `rrm/afc/failed` and a Slack alert is dispatched. The `del_afc_info()` function atomically clears entries from both `rrm/afc/aps` and `rrm/afc/failed`, ensuring no zombie entries persist across claim cycles.

4.13 Multi-Radio AP Band Selection

Figure 15

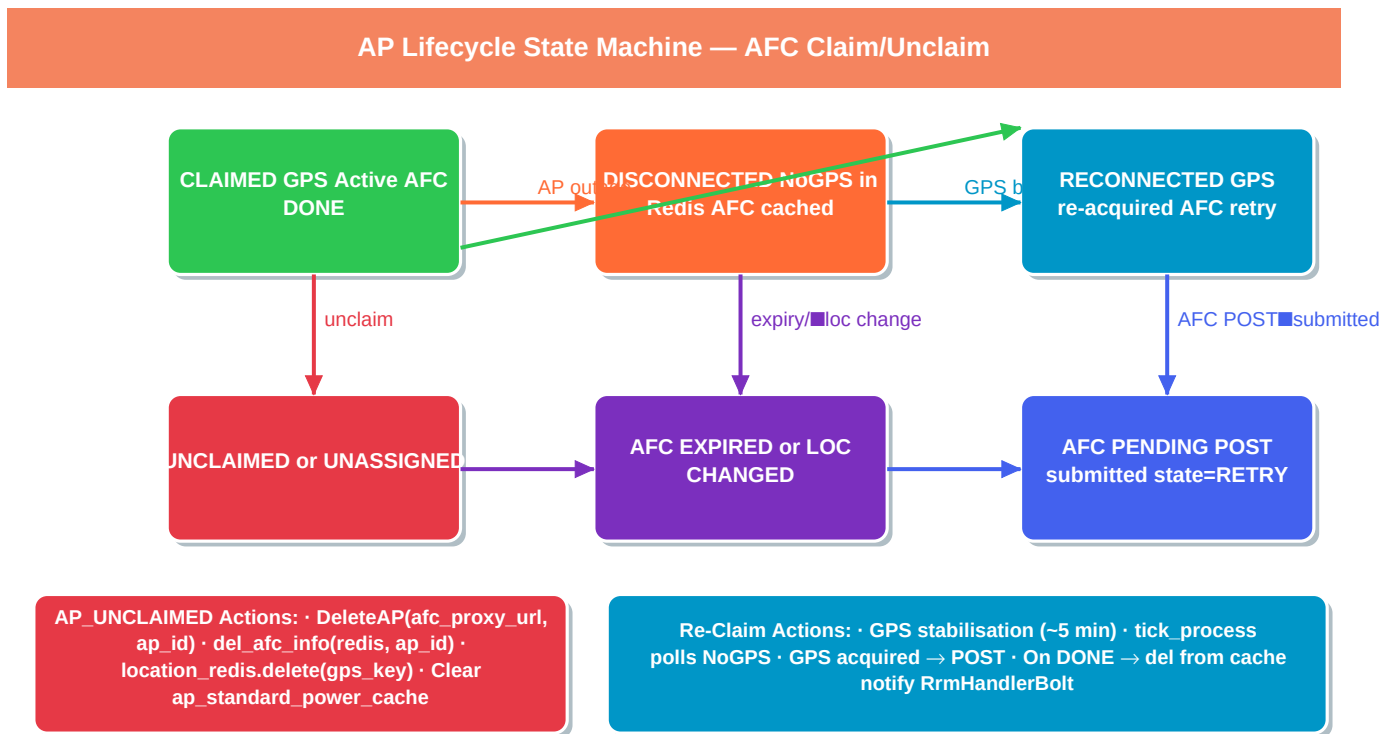


Fig. 15 — AP Lifecycle State Machine — AFC Claim/Unclaim Workflow

The `updateGraphForAFC()` function in `mistGraph.py` applies AFC enrichment exclusively to band-6 RRM graphs. For each node where `afc_standard_power=True`, it invokes `afcNode()` to attach AFC channels and EIRP constraints. The regex pattern `re.match(r'[a-f0-9]{12}:r24', node)` identifies 2.4 GHz radios being converted to band-6 operation. For such nodes, `auto_band=True` is set and the `rrm_channel` is re-selected from the AFC-authorized channel set to ensure the converted radio immediately operates on a compliant channel.

4.14 Multi-Provider Dynamic Selection

Figure 16

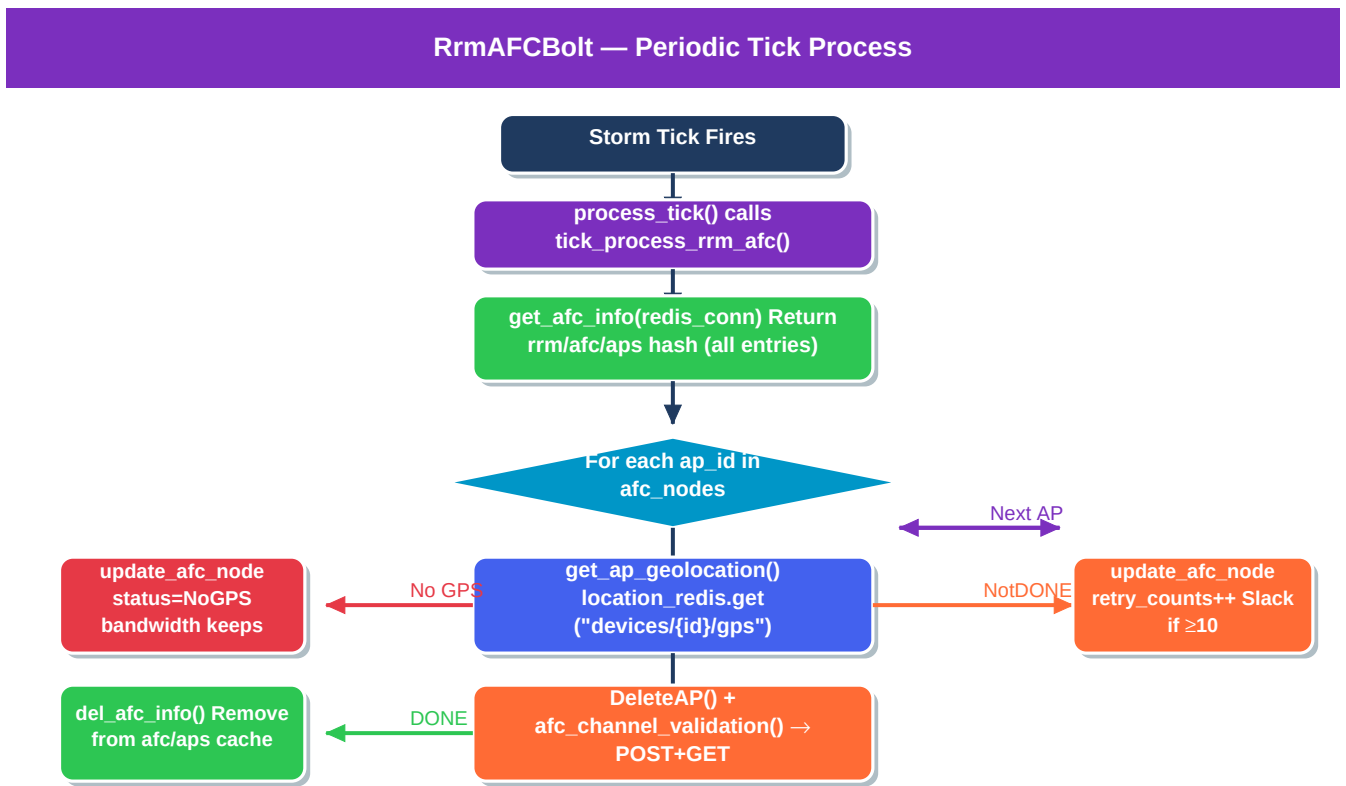


Fig. 16 — RrmAFCBolt Periodic Tick Process (*tick_process_rrm_afc()*)

The provider selection mechanism is entirely runtime-configurable without firmware or code changes. The `afc-provider-flag` RRM policy parameter accepts string values '0', '1', '2', or 'dev'. The `get_flag_provider_mapping()` function returns the corresponding `providerName` string that is embedded in the AFC request payload. The AFC proxy uses this `providerName` to route the request to the appropriate upstream provider.

In development mode (`flag='dev'`), the system never contacts an external provider. Instead, it checks Redis for an operator-injected mock AFC response (`get_mock_afc_response()`), enabling controlled testing of specific channel assignments and EIRP values without requiring a live AFC system.

4.15 AP Lifecycle State Machine

Figure 17

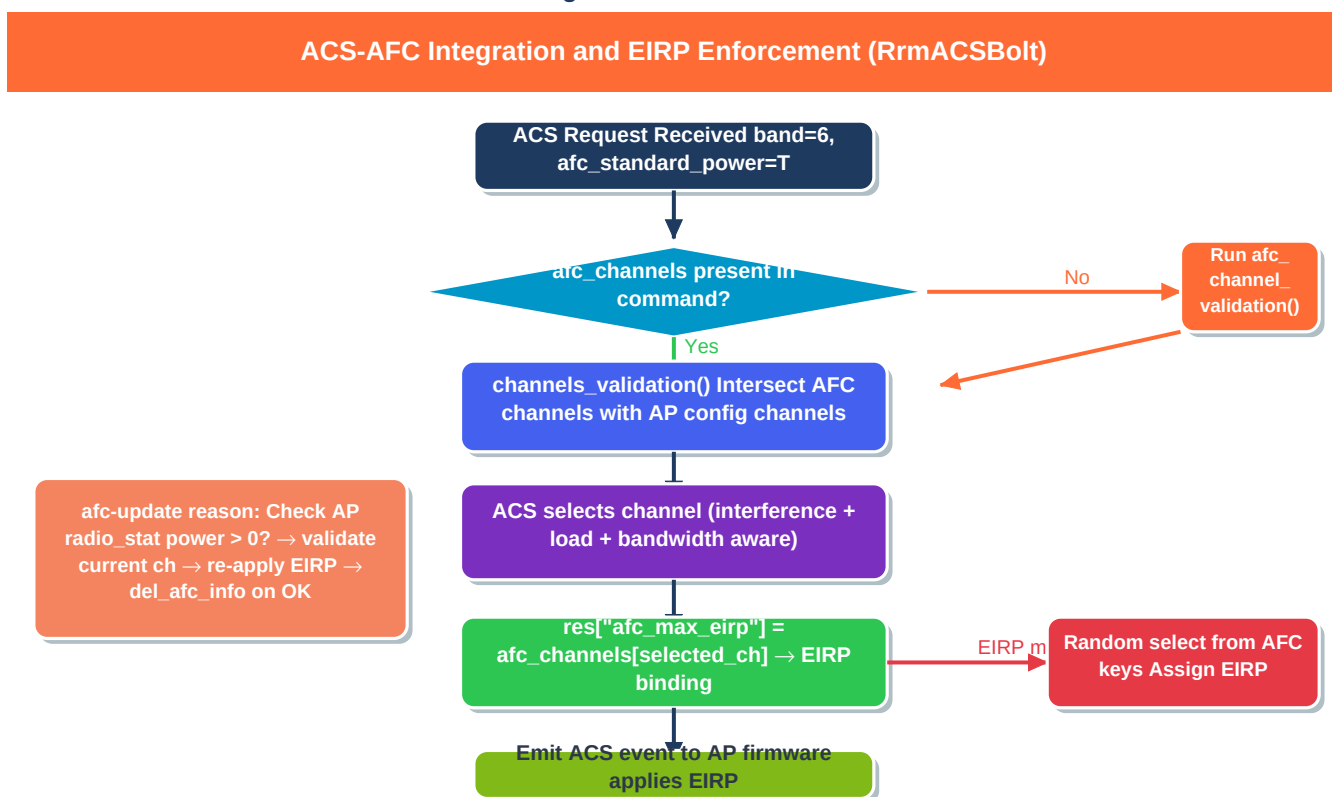


Fig. 17 — ACS-AFC Integration and EIRP Enforcement (RrmACSBolt.process_tup())

The AP lifecycle state machine governs AFC registration state across the full AP provisioning lifecycle. The RrmAFCBolt maintains an `ap_standard_power_cache` dictionary that records the standard-power capability of each AP along with its `org_id` and `site_id`. This cache enables the bolt to populate site and org context in lifecycle events even when the incoming event tuple lacks this information.

The `process()` method handles format compatibility between legacy (string status) and current (dictionary with `ap_status`, `site_id`, `org_id`) event formats, ensuring backward compatibility across rolling firmware upgrades. When `AP_UNASSIGNED` or `AP_UNCLAIMED` status is detected, the `filtered_ap_map` is passed to `ap_event_handler()` for atomic cleanup.

4.16 RrmAFCBolt Tick Process

Figure 18

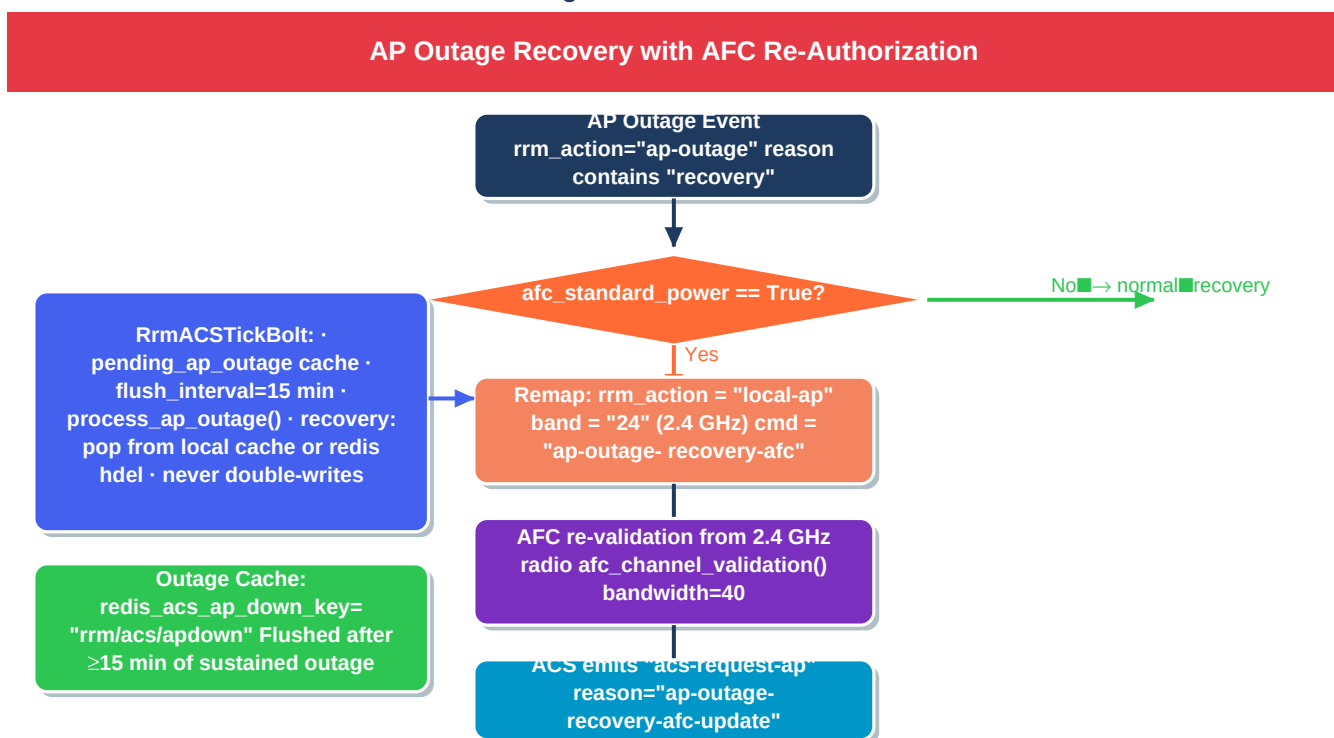


Fig. 18 — AP Outage Recovery with AFC Re-Authorization (ap-outage-recovery-afc)

The `process_tick()` method in `RrmAFCBolt` calls `tick_process_rrm_afc()` on every Apache Storm tick tuple. This function polls the `rrm/afc/aps` Redis hash and processes each entry in sequence. For each AP in the retry queue, it: (1) retrieves `org_id`, `site_id`, and `bandwidth` from the stored JSON; (2) calls `get_ap_geolocation()` to check for GPS availability; (3) if GPS is present, fetches the per-site RRM policy to obtain `provider_flag` and `location_threshold`; (4) calls `DeleteAP()` to clear any stale proxy entry; (5) calls `afc_channel_validation()` to obtain fresh authorisation; and (6) updates Redis with the new status or removes the entry on success.

4.17 ACS-AFC Integration and EIRP Enforcement

`RrmACSBolt.process_tup()` implements the critical integration between ACS and AFC. The `avoid_frequent_update()` guard is bypassed for AFC-specific commands ('`afc-update`', '`afc-channel-update`', '`ap-outage-recovery`') to ensure these high-priority operations are never throttled. The '`afc-update`' reason path additionally checks the AP's current radio power status (`radio_stat.power > 0`) before applying new channel/EIRP constraints, preventing unnecessary channel changes for offline radios.

4.18 AP Outage Recovery with AFC Re-Authorisation

The `RrmACSTickBolt` implements a hysteresis mechanism for AP outage events: outage events are held in an in-memory `pending_ap_outage` cache for 15 minutes before being written to Redis. If an

AP recovers within this window, the outage entry is dropped locally without any Redis write. This prevents transient power events from triggering unnecessary channel changes. Only sustained outages (>15 minutes) result in Redis state changes and subsequent AFC re-authorisation on recovery.

4. DETAILED DESCRIPTION (continued)

4.19 AFC Response Schema and State Handling

Figure 19

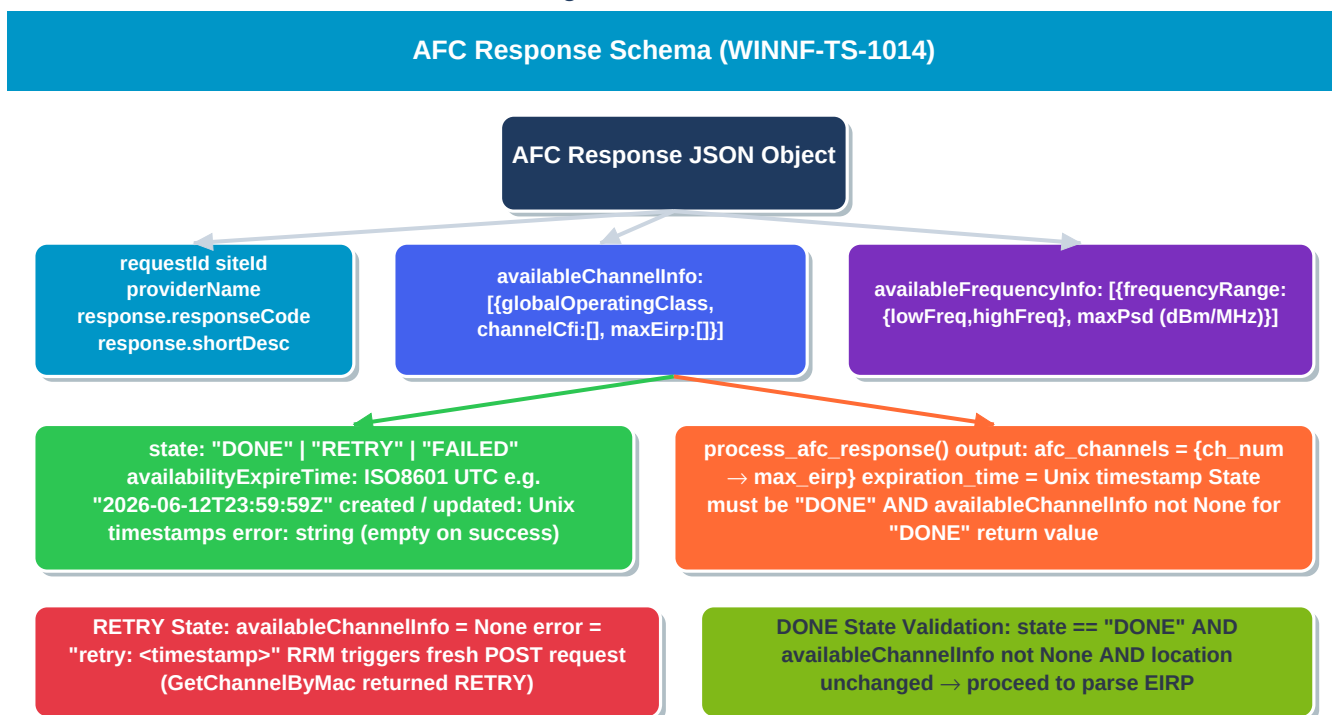


Fig. 19 — AFC Response Schema and State Handling (WINNF-TS-1014)

The AFC response schema follows WINNF-TS-1014 v1.4. The state field governs RRM behaviour: DONE with non-null availableChannelInfo triggers immediate process_afc_response(); RETRY with null channels triggers a fresh POST (the error field contains 'retry: '); FAILED triggers rrm/afc/failed recording.

Figure 20

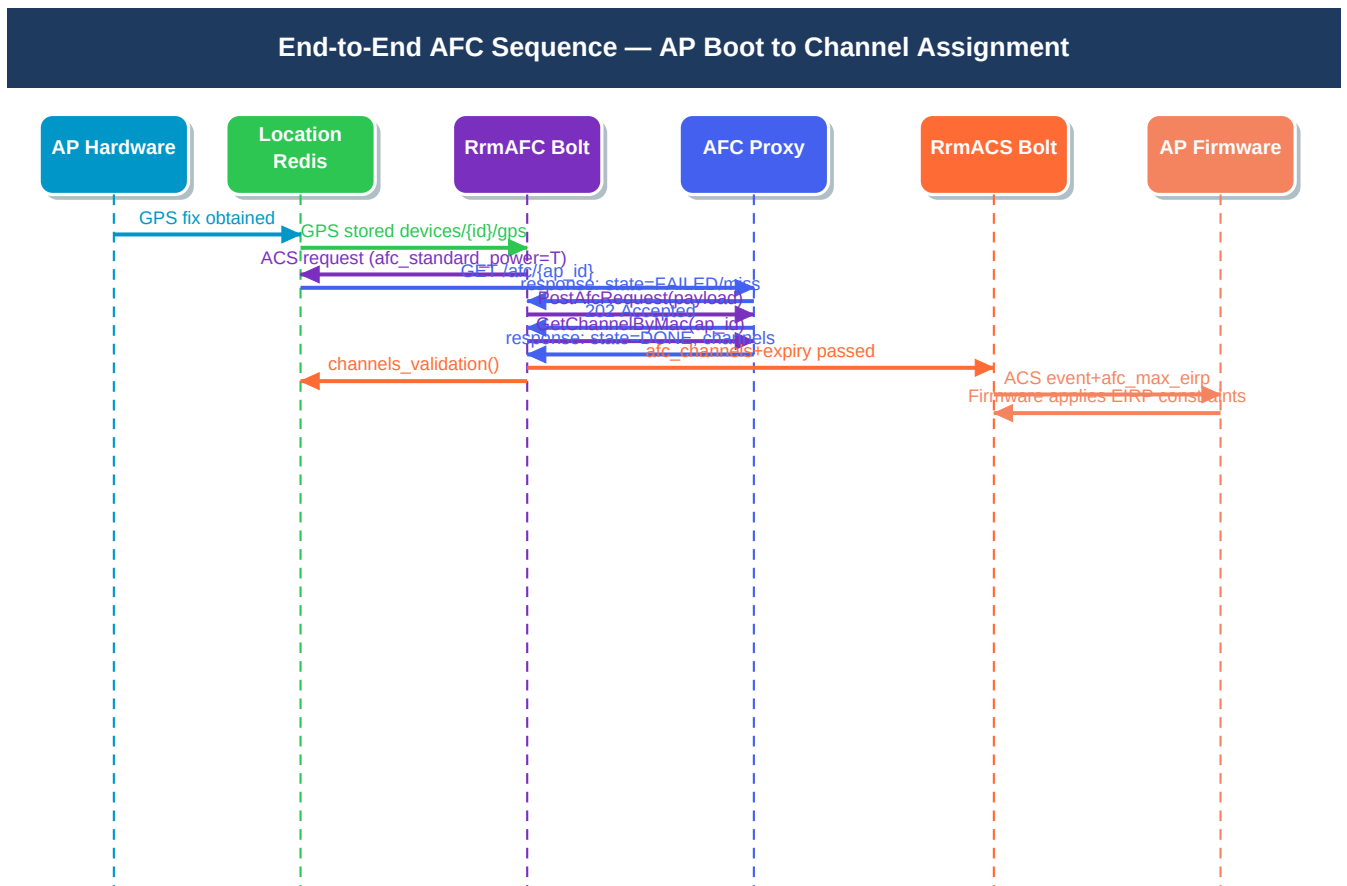


Fig. 20 — End-to-End AFC Sequence — AP Boot to Channel Assignment

4.20 End-to-End AFC Sequence

The complete AP boot-to-channel-assignment sequence begins with GPS fix acquisition and Location Redis update, proceeds through RRM ACS request, AFC GET/POST/GET cycle, channel and EIRP assignment, and concludes with AP firmware applying the EIRP constraints as: $\text{tx_power} = \text{afc_max_eirp} - \text{antenna_gain} - 10 \cdot \log_{10}(\text{chains})$.

Figure 21

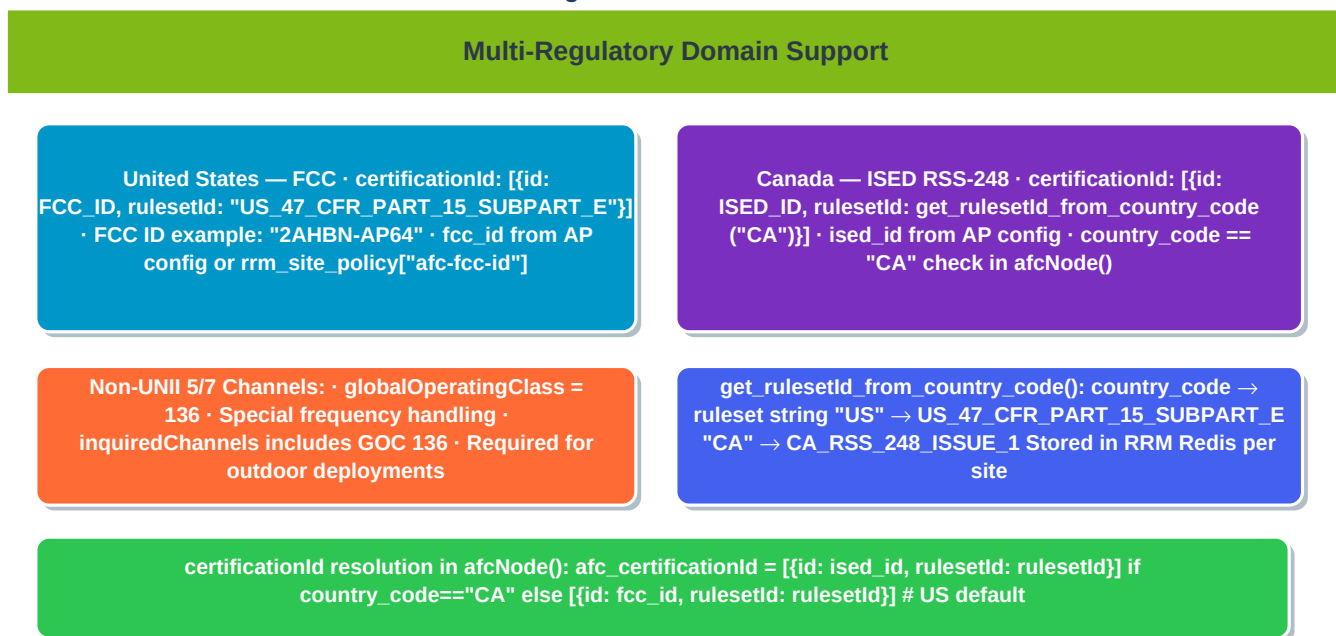


Fig. 21 — Multi-Regulatory Domain Support (US FCC + Canada ISED)

4.21 Multi-Regulatory Domain Support

Country-specific regulatory compliance is achieved through dynamic `certificationId` construction in `afcNode()`. The `get_rulesetId_from_country_code()` function maps the AP's `country_code` (from the RRM graph's `country_code` attribute) to the appropriate ruleset string. For US (`country_code='US'`), the `rulesetId` is 'US_47_CFR_PART_15_SUBPART_E'. For Canada (`country_code='CA'`), both the ISED ID and the Canadian ruleset are used. The Non-UNII 5/7 channels (`globalOperatingClass 136`) are included in the `inquiredChannels` array to enable outdoor deployments requiring access to upper 6 GHz spectrum.

Figure 22

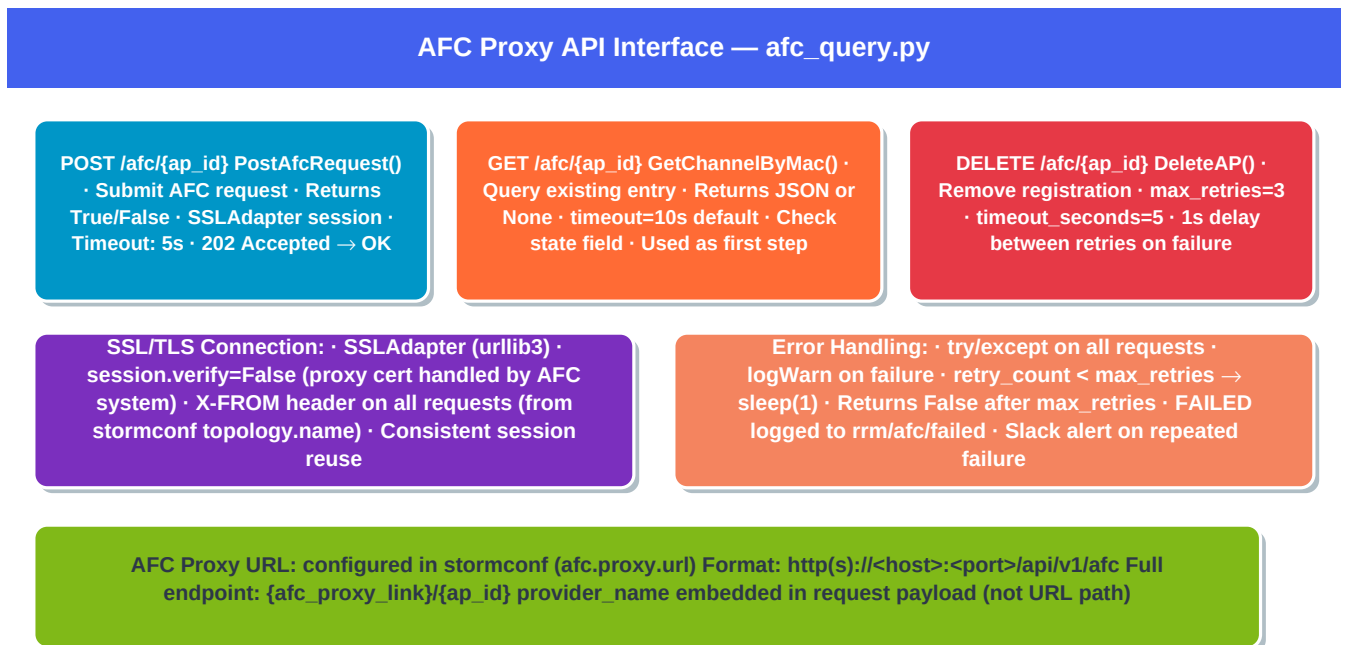


Fig. 22 — AFC Proxy API Interface — POST/GET/DELETE Operations (afc_query.py)

4.22 AFC Proxy API Interface

All AFC proxy communication uses HTTPS with a custom SSLAdapter (SSLAdapter inheriting from HTTPAdapter) that configures the SSL context appropriately for the AFC proxy's certificate. The `session.verify=False` setting disables server certificate verification, as the AFC proxy's certificate chain is managed by the AFC system operator rather than a public CA. An X-FROM header identifying the Storm topology name is included in all requests for audit and debugging purposes.

Figure 23

AFC Test Scenario Coverage Matrix		
test_ap_events	AP_UNCLAIMED/AP_UNASSIGNED	DeleteAP+del_afc_info+GPS clear
test_afc_tick_process	NoGPS→DONE transition	tick_process_rrm_afc with GPS
test_afc_tick_location_threshold	Location change detection	ECEF >50000m threshold
test_rrm_acs_nogps	ACS with no GPS	NoGPS recorded, radio disabled
test_rrm_global_afc_nogps_lpi_fallback	LPI fallback when no GPS	Non-SP ACS event emitted
test_rrm_global_afc_gps_standard_power	Full SP flow with GPS	DONE, channels assigned
test_afc_RETRY_state	Provider returns RETRY	POST resubmit triggered
test_afc_location_changed	AP relocated > threshold	DELETE+POST+GET cycle

Fig. 23 — AFC Test Scenario Coverage Matrix (mistsys/mist-rrm-exp)

Figure 24

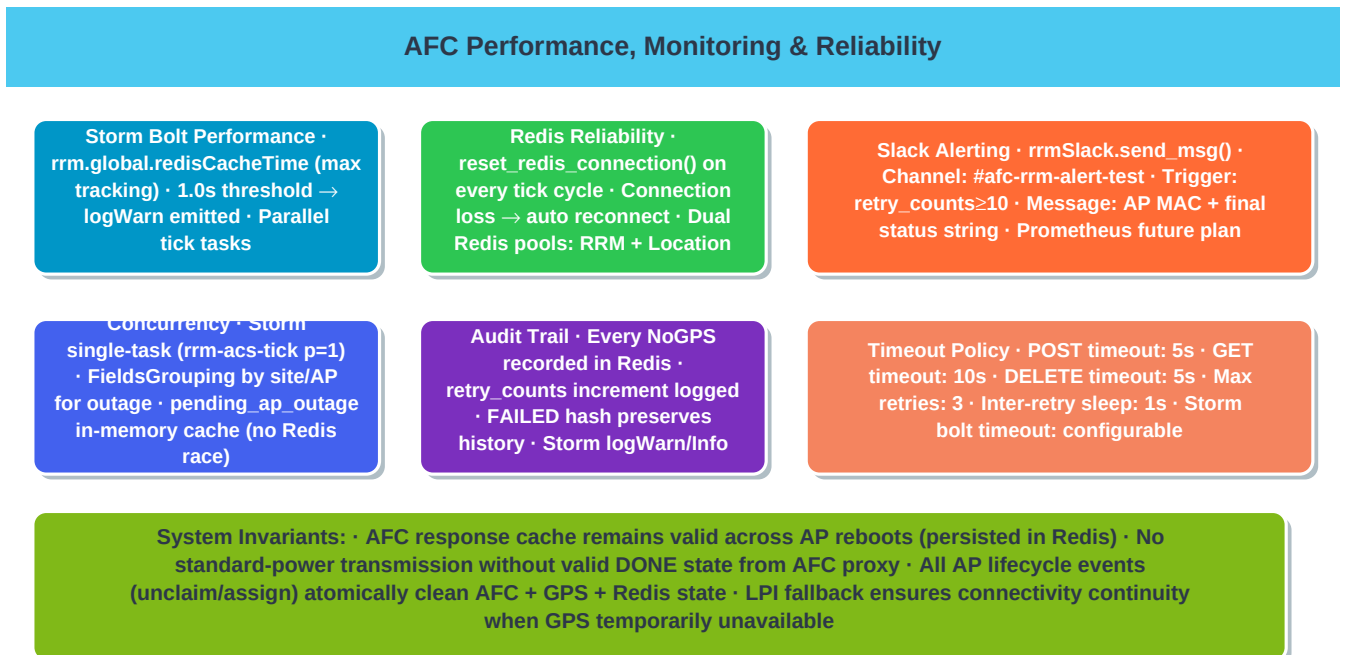


Fig. 24 — AFC Performance, Monitoring, and Reliability Architecture

5. PATENT CLAIMS

Independent Claims

Claim 1: A system for Automated Frequency Coordination (AFC) lifecycle management in a wireless network comprising: a Radio Resource Management (RRM) controller configured to construct AFC request payloads including device certification identifiers resolved per access point (AP) model and regulatory domain; a geolocation validation engine employing Earth-Centered Earth-Fixed (ECEF) coordinate transformation to detect AP displacement beyond a configurable threshold; a two-phase GET-before-POST AFC query protocol minimising provider load while ensuring response freshness and location integrity; a multi-provider selection module configurable per site via RRM policy without firmware changes; a cache reuse engine implementing a three-tier expiry policy based on time-to-expiry and location consistency; and a Power Spectral Density (PSD)-to-EIRP conversion engine enforcing per-bandwidth AFC minimum PSD floors.

Claim 2: A method for RRM-driven AFC lifecycle management comprising: determining radio eligibility for AFC based on AP model, band capability, and operational state; constructing an AFC request payload conformant to WINNF-TS-1014, including GPS-derived location ellipse parameters, ruleset identifier, and certification identifier per regulatory domain; querying an AFC proxy via HTTP GET to check for an existing valid entry; upon detection of location mismatch via ECEF coordinate comparison, atomically deleting the stale proxy entry and resubmitting via HTTP POST; parsing the AFC response to extract per-channel EIRP as the minimum of maxEirp and PSD-derived EIRP, floored by bandwidth-specific AFC_MIN_PSD; and applying the computed EIRP to the ACS-selected channel before AP firmware transmission.

Claim 3: A system for AFC NoGPS resilience comprising: a GPS unavailability detection module that records NoGPS conditions in a persistent Redis cache with retry metadata including org_id, site_id, bandwidth, timestamp, and retry_counts; a Low Power Indoor (LPI) fallback module that determines AP model LPI capability and emits a non-standard-power ACS event maintaining client connectivity when GPS is unavailable; a periodic tick-driven retry module that polls the NoGPS Redis queue and re-attempts AFC validation upon GPS availability; and an escalation module that moves APs to a failed cache and issues external alerts after a configurable maximum retry count.

Claim 4: A system for 320 MHz Wi-Fi 7 AFC channel mode disambiguation comprising: a mode classification function that assigns each 320 MHz channel to Mode 1 (primary sub-channel PSC $\in \{31, 95, 159\}$) or Mode 2 (PSC $\in \{63, 127, 191\}$) based on its inclusion in the respective pre-defined sub-channel range; a per-mode EIRP expansion engine that expands EIRP values from AFC responses only to sub-channels within the same mode as the PSC; and a mode consistency validator that rejects EIRP assignments to sub-channels belonging to a different mode, preventing cross-mode power limit violations.

Claim 5: A method for radio-aware AFC request origination in multi-radio access points comprising: determining, by RRM, whether the AP contains a dedicated 6 GHz radio interface; when a dedicated 6 GHz radio is absent but a 2.4 GHz radio with auto-band capability is present, originating the AFC request from the 2.4 GHz radio interface with `auto_band=True` and `afc_standard_power=True`; re-selecting the RRM channel from the AFC-authorized channel set after band conversion; and propagating standard-power flags to the converted radio to ensure post-conversion EIRP compliance.

Dependent Claims

Claim 6: The system of Claim 1, wherein the certification identifier is selected from an FCC ID for United States regulatory domain with `rulesetId='US_47_CFR_PART_15_SUBPART_E'`, or an ISED ID for Canadian regulatory domain with `rulesetId` resolved via `get_rulesetId_from_country_code('CA')`, based on the AP's configured country code.

Claim 7: The method of Claim 2, wherein the location ellipse orientation is normalised by subtracting 180 degrees when the GPS-reported orientation exceeds 180 degrees, preserving geometric equivalence while satisfying AFC system input validation constraints.

Claim 8: The system of Claim 1, wherein the multi-provider selection module maps policy flag values to provider names: '0'→'wfa' (production), '1'→'wfa-th01' (internal test harness), '2'→'wfa-th02' (certification harness), and 'dev'→'dev' (development mock using Redis-injected or synthesised responses).

Claim 9: The method of Claim 2, wherein upon detection of an `AP_UNCLAIMED` or `AP_UNASSIGNED` lifecycle event: the AP registration is deleted from the AFC proxy via HTTP DELETE with up to three retry attempts and a five-second timeout per attempt; the AP's AFC information is removed from both active and failed Redis hash stores; and the AP's GPS location cache is deleted from the Location Redis service.

Claim 10: The system of Claim 1, wherein the three-tier cache policy comprises: Tier 1 reusing cached `afc_channels` when time-to-expiry exceeds two hours without issuing any AFC query; Tier 2 issuing a GET validation when time-to-expiry is between zero and two hours; and Tier 3 performing a full DELETE-POST-GET refresh cycle when expired or when location change is detected, regardless of remaining expiry time.

Claim 11: The method of Claim 2, wherein the maximum AFC response validity period is capped at sixteen hours by the RRM controller, independently of the AFC provider's stated `availabilityExpireTime`, requiring re-registration upon expiry.

Claim 12: The system of Claim 3, wherein the periodic retry module applies per-site configurable provider flags and location thresholds retrieved from the RRM policy service, and executes `DeleteAP()` before `afc_channel_validation()` to prevent duplicate registration conflicts at the AFC proxy.

Claim 13: The method of Claim 2, wherein when the ACS-selected channel is absent from the AFC-authorized channel map, a replacement channel is selected using a random-choice algorithm from the set of AFC-authorized channels, and the corresponding EIRP from the AFC-authorized map is assigned to the result.

Claim 14: The system of Claim 1, further comprising an AP outage recovery module that, upon detecting an AP radio outage recovery event for a standard-power AP, remaps the recovery action to originate from the 2.4 GHz radio interface, performs fresh AFC re-validation, and emits an `acs-request-ap` command with `reason='ap-outage-recovery-afc-update'` to apply the new authorisation.

Claim 15: The method of Claim 2, wherein the GET-before-POST protocol additionally handles the RETRY state by issuing a fresh POST request and subsequent GET validation, treating the RETRY state as equivalent to a cache miss for the purpose of triggering new provider registration.

Claim 16: The system of Claim 1, further comprising an AP outage hysteresis module that maintains outage events in memory for a configurable flush interval of fifteen minutes before persisting to Redis, allowing transient outage-recovery pairs to be dropped without triggering AFC re-authorisation cycles.

Claim 17: The method of Claim 5, wherein the 2.4 GHz radio conversion to 6 GHz operation is identified by matching the radio identifier against the pattern `[a-f0-9]{12}:r24`, and the converted radio's RRM channel is re-selected from the AFC-authorized channel set to ensure immediate compliance post-conversion.

Claim 18: The system of Claim 4, wherein the PSD-to-EIRP conversion computes `AFC_MIN_PSD`-floored per-channel EIRP for all sub-channels within a 320 MHz channel using bandwidth-specific delta arrays $\delta = [\pm 2, \pm 6, \pm 10, \pm 14, \pm 18, \pm 22, \pm 26, \pm 30]$, applied exclusively to sub-channels within the same Mode 1 or Mode 2 group as the primary sub-channel.

Claim 19: The method of Claim 2, wherein the AFC request payload includes Non-UNII-5 and Non-UNII-7 channels via `globalOperatingClass=136` in the `inquiredChannels` array, enabling outdoor Standard Power operation across the full regulatory allocation including upper 6 GHz spectrum.

Claim 20: The system of Claim 1, wherein connection reliability is maintained through: dual independent Redis connection pools (RRM Redis and Location Redis) with automatic reset on each periodic tick; per-request connection timeout policies (POST: 5s, GET: 10s, DELETE: 5s); and a maximum of three retry attempts with one-second inter-retry delays for all AFC proxy operations.

Claim 21: The method of Claim 2, wherein the `afc_max_eirp` value assigned to the ACS-selected channel is passed downstream to the AP firmware for application as: `tx_power`

= $\text{afc_max_eirp} - \text{antenna_gain} - 10 \cdot \log_{10}(\text{radio_chains})$, ensuring the radiated power at the antenna port complies with the per-channel EIRP limit.

Claim 22: The system of Claim 3, wherein the NoGPS condition is persisted to Redis for audit purposes regardless of whether LPI fallback is invoked, recording {ap_id, ap_status, bandwidth, timestamp, retry_counts} to enable post-incident analysis and compliance verification.

Claim 23: The method of Claim 2, further comprising constructing the location ellipse uncertainty parameters by capping majorAxis and minorAxis to a maximum of 325 metres per WINNF-TS-1014 constraints, and normalising the height value to a minimum of 0.1 metres Above Ground Level (AGL) to satisfy AFC system input validation.

Claim 24: A non-transitory computer-readable medium storing instructions that, when executed by a processor, implement the method of any of Claims 2, 5, or 9 in an Apache Storm distributed stream processing topology, wherein AFC lifecycle management is performed within named Storm bolts communicating via typed streams.

Claim 25: The system of Claim 1, wherein the RRM graph enrichment function updateGraphForAFC() selectively applies AFC channel and EIRP constraints only to 6 GHz band graph nodes where `afc_standard_power=True`, leaving Low Power Indoor nodes, 2.4 GHz nodes, and 5 GHz nodes unaffected by AFC authorisation status.

6. PRIOR ART COMPARISON

The following table distinguishes the present invention from known prior AFC implementations:

Feature	Prior Art	This Invention
AFC Request Origin	Fixed per AP; firmware-embedded	RRM-constructed; model-specific FCC/ISED IDs
Query Protocol	POST always; no cache check	GET-first; POST only on miss/change/RETRY
Location Validation	Lat/lon delta (geographically inconsistent)	ECEF Cartesian distance (geographically uniform, metres)
Provider Selection	Static configuration	Per-site RRM policy; runtime flag mapping
NoGPS Handling	Disable 6 GHz radio immediately	LPI fallback + Redis retry queue + tick retry
320 MHz Mode	Not addressed	Mode 1/Mode 2 PSC disambiguation (get_mode())
EIRP Computation	maxEirp only	$\min(\text{maxEirp}, \text{psdEirp}) + \text{AFC_MIN_PSD floor}$
Cache Expiry	Binary valid/expired	3-tier: reuse / validate / refresh (with 16h cap)
Claim/Unclaim	Manual; proxy not updated	Atomic DELETE + Redis clear + GPS cache clear
AP Outage Recovery	Standard recovery; no AFC re-auth	ap-outage-recovery-afc: fresh 2.4GHz re-validation
Multi-Regulatory	US FCC only	US (FCC) + CA (ISED) + extensible via country_code
ACS-AFC Integration	Separate systems; no EIRP binding	Inline EIRP enforcement: afc_max_eirp post-ACS
Radio Origination	Dedicated 6 GHz only	2.4 GHz origination for auto-band / tri-band APs
Retry Management	Simple retry; no audit	retry_counts, failed hash, Slack alerting
Test Coverage	Device-level only	Full unit/integration test suite with mock AFC

7. IMPLEMENTATION REFERENCE

The invention is implemented and evidenced in the mistsys/mist-rrm-exp repository. The following source modules constitute the reference implementation:

Module	Path	Inventive Contribution
rrmAFC.py	src/rrmAFC.py	RrmAFCBolt: AP lifecycle, claim/unclaim, tick retry, Slack alert
afc_utils.py	src/afc/afc_utils.py	afc_channel_validation, ECEF location, NoGPS, tick_process_rrm_afc
afc_payload.py	src/afc/afc_payload.py	get_payload_proxy, channel-freq mapping, 320MHz mode, PSD defaults
afc_query.py	src/afc/afc_query.py	PostAfcRequest, GetChannelByMac, DeleteAP with retry logic
APnode.py	src/common/APnode.py	afcNode: 3-tier cache, LPI fallback, graph node enrichment
mistGraph.py	src/common/mistGraph.py	updateGraphForAFC: band-6 enrichment, auto_band conversion
rrmHandler.py	src/rrmHandler.py	rrm_afc_channels: AFC proxy response routing to ACS stream
rrmACS.py	src/rrmACS.py	ACS + AFC EIRP enforcement, outage recovery, afc-update path
rrmACSV2.py	src/rrmACSV2.py	Enhanced ACS v2: inline AFC validation, afc_max_eirp binding
rrmRedis.py	src/common/rrmRedis.py	set_new/update/del/get_afc_node, retry_counts, failed hash
rrmACSTick.py	src/rrmACSTick.py	AP outage hysteresis: 15-min cache, pending_ap_outage
test_rrm_afc.py	src/tests/	Unit tests: AP events, tick process, location threshold, RETRY
test_rrm_global_afc_lpi.py	src/tests/	LPI fallback, standard-power GPS flow integration tests

8. INVENTORS

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